SmartLib: A Web and Mobile-Based Smart Library System with Online Printing Service Requests for STI College Ormoc

A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Ormoc

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology

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May 21, 2026

ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **SmartLib: A Web and Mobile-Based Smart Library System with Online Printing Service Requests for STI College Ormoc**

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In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.

ENDORSED BY:

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MAY 21, 2026

APPROVAL SHEET

This capstone project proposal titled **SmartLib: A Web and Mobile-Based Smart Library System with Online Printing Service Requests for STI College Ormoc**, prepared and submitted by **John Bazty C. Cantay, Keith Gaudi Gerard P. Juntilla**, **Buentheo Nathaniel A. Noval,** in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

Elizabeth L. Dumaran
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Accepted and approved by the Capstone Project Review Panel
in partial fulfillment of the requirements for the degree of
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INTRODUCTION

Project Context

In the modern educational landscape, the library remains a central hub for academic activity; however, the global shift toward digitalization has redefined how students interact with information. Historically, libraries were managed through manual ledgers and index cards, but as technology evolved, the concept of a "Smart Library" emerged to address modern efficiency needs. Despite these trends, many regional institutions in the Philippines, including STI College Ormoc, continue to operate with manual systems that fail to meet the fast-paced demands of 21st-century learners.

The prevailing problem situation at STI College Ormoc is characterized by a heavy reliance on manual labor. Currently, automated circulation is non-existent. The "borrow and return" cycle requires students to manually sign index cards, while librarians must hand-write transaction details. This approach makes inventory management a significant challenge, as tracking the physical condition and location of resources requires constant manual auditing. Monitoring overdue materials presents an additional challenge. In the absence of automated notifications, librarians must manually review index cards to identify outstanding loans. This delayed tracking complicates penalty assessments and leads to missing resources, which leaves students without essential learning materials.

Handling printing manually makes it harder for the library to run smoothly, often causing crowded desks and slow service. Students must physically go to the library just to submit a file for printing, it creates unnecessary crowds and wastes a lot of time. This manual printing workflow requires students to submit their files physically via USB, Email, or Messenger, after which the librarian must manually check file formats and page counts, set print options, calculate costs, collect cash payments, monitor the queue, check the printed output, and physically record the transaction details in a traditional paper logbook.

The current manual log-in process forces users to wait in lines to fill out physical paper attendance forms. Users must handwrite details including the current date, time-in, full name, educational level (Senior High School or College), course/strand, section, and their specific reason for visiting (Library Visit, Book Borrowing, or Printing). This is followed by a mandatory signature and physical verification by the librarian before the student is allowed to enter the library premises.

This project is justified by the need to bridge the gap between traditional manual resources and modern digital expectations. By addressing these unresolved elements, the library can transition from a place of paperwork to a more efficient study environment. Therefore, there is a clear relationship between the need for operational modernization and the development of a localized smart management system.

Purpose and Description

The primary purpose of the SmartLib system is to modernize and centralize institutional library services into a single, cohesive web and mobile platform. The system streamlines administrative workflows for both librarians and students, replacing labor-intensive manual methodologies. Its primary objective is to optimize book search, borrowing, and return transactions, while introducing self-service modules that minimize transaction overhead. By automating routine administrative tasks, the platform reduces the staff's manual burden, allowing the library to operate as an efficient study and research environment.

What makes this project unique and innovative is how it tackles the specific, real-world problems faced by students on a daily basis. Beyond just tracking books, the system introduces a smart printing request feature. This allows students to handle their printing needs from their own devices instead of having to manually go to the library and hand over the files via Gmail or USB. Additionally, the project is relevant because it respects the physical space of the library. By utilizing QR codes for check-in, the system offers a seamless experience and ensures precise attendance logging. This combination of traditional library tools and modern digital services makes the system a truly "smart" solution tailored to the actual needs of the campus community.

The project is also designed to be highly engaging and helpful for the modern learner. It includes a personalized dashboard where students can see exactly what they have borrowed and when they need to return it, helping them stay organized and avoid penalties. Furthermore, it adds a touch of innovation through a recommendation feature that suggests books based on what is popular or relevant to a student's specific course. By making the library's inventory more visible and accessible through a digital catalog, the system ensures that the school's resources are used to their full potential. This project is a relevant step forward, turning a traditional facility into a modern, efficient, and student-centered academic hub.

Objectives

General Objective

To design, develop, and implement SmartLib that automates traditional library operations, including circulation, inventory tracking, and student services, to enhance the overall academic experience at STI College Ormoc.

Specific Objectives

- To develop a Digital Cataloging interface that enables users to search for books and research papers with real-time availability status and physical shelf location tracking.
- To create an Automated Circulation module using barcode scanning for physical books and dynamic, time-synchronized QR code verification for student identification and check-out processing to prevent identity fraud.
- To build an Inventory Management system that allows librarians to monitor resource condition, perform bulk data imports of legacy manual records, and generate automated inventory reports.
- To implement a contactless, dynamic QR-based attendance monitoring feature that tracks student entry and exit, requires users to define their specific purpose of visit, and identifies peak library usage times.
- To integrate an Online Printing Service Request system allowing students to upload

files in restricted secure formats (e.g., PDF, DOCX), monitor real-time printer availability status, and schedule pickup times to minimize counter queues.

- To deploy an interactive tablet-based Library Kiosk within the premises to facilitate rapid local catalog search, shelf location discovery, and automated academic citation generation.
- To provide an Admin Analytics Dashboard for library staff to view insights on the most borrowed resources and student usage trends.
- To develop a Fines, Violations, and Accounts Clearance module that automatically computes overdue book penalties based on customizable operating hours and holiday exclusion rules, records non-circulation behavioral infractions (e.g., noise, food violations), applies maximum penalty caps, monitors outstanding student balances, and supports both manual cash clearance at the front desk and semi-automated verification via GCash reference validation.
- To build a Printing Service Management and Consumables Tracking subsystem for library staff to manage print queues, monitor revenue, and track real-time supplies such as ink and paper stock.

Scope and Limitations

Scope

The project focuses on the development of a multi-platform system comprising a web-based administrative dashboard, a dedicated React Native mobile application for student and faculty access, and a tablet-based local kiosk application for catalog search and shelf tracking within the library premises. The primary users are the students and faculty members of STI College Ormoc, who will utilize the digital catalog, mobile app, kiosk, and printing services, and the library staff, who will manage the inventory, attendance, and circulation. This research and development project is conducted during the second semester of the 2025–2026 academic year.

- Digital Cataloging: A searchable database for books and research papers with real-time availability status and physical shelf coordinates.

- Automated Circulation: A system utilizing physical barcode scanning for books and dynamic student QR codes for secure check-in and check-out.

- Inventory Management: Automated tracking of book conditions, new arrivals, and stock levels for administrative use.

- QR-Based Attendance Monitoring: A dynamic contactless entry and exit tracking module requiring visitors to specify their purpose of visit (e.g., study, printing, borrowing).

- Online Printing Service Request: A module allowing students to upload files in restricted secure formats (e.g., PDF, DOCX), monitor real-time printer availability status, and schedule pickup times to minimize counter queues.

- Interactive Library Kiosk: A local tablet interface for self-service book queries, physical shelf location tracking, and APA/MLA academic citation generation.

- Admin Analytics: A dashboard providing insights into peak usage hours and popular library resources.

- Reports Generation: An administrative module capable of generating exportable and printable reports on inventory audits, daily attendance statistics, printing transactions, and outstanding student penalty fines for unreturned books.

Limitations

- The system is specifically designed for the library of STI College Ormoc and may not be fully compatible with the infrastructure of other STI campuses without modification.

- The system requires a continuous internet connection to operate, although the mobile application supports time-synchronized offline QR code generation for entry logging fallback.

- The Online Printing Service Request and Fines modules exclude automated merchant payment gateway APIs, relying instead on library front-desk cash collections and manual GCash receipt verification queues, and do not handle physical hardware maintenance.
- The system operates independently and does not live-sync with other campus databases (e.g., accounting or registrar).

Review of Related Literature/Studies/Systems

The rapid digital transformation of academic libraries has led to the development of smart library management systems that integrate cataloging, circulation, user services, and analytics into unified platforms. Recent works on web-based and intelligent library management systems emphasize replacing manual, paper-based processes with automated modules for book transactions, member registration, and inventory control to improve accuracy and efficiency in service delivery (Chen & Ahmed, 2024; Kumar, 2023). In line with this shift, digital cataloging, automated circulation, attendance monitoring, and online service requests have become central themes in contemporary library technology research, providing the conceptual and technical foundation for the proposed STI College Ormoc SmartLib.

Local Institutional System Developments

A relevant local study for this project is the STI College Ormoc Online Library System with Electronic Access for Reservation and SMS Notification, developed by Gormano and Romo (2017). Their system was built using a WAMP server with a PHP backend and used JavaScript and AJAX to keep the user interface smooth. It aimed to move the library away from paper logbooks by digitizing basic records (like book profiles, categories, and borrower details), tracking daily transactions (borrowing, returning, and reserving), and sending out automated text reminders. While their project proved that digital reservations and notifications could work on campus, it only handled basic circulation and relied on extra SMS hardware to send texts. The proposed SmartLib system builds directly on what they started but upgrades it into a dual-platform web and mobile-responsive system. By adding practical, low-cost features that the 2017 version lacked, such as QR codes for quick borrowing and attendance, an online printing queue, live analytics dashboards, and downloadable admin reports, this new project fills the remaining gaps to create a truly modern study environment for the school.

Digital cataloging and OPAC services

Digital cataloging techniques have evolved from physical index cards to metadata-driven, searchable databases that provide real-time resource availability (Kumar, 2023; Singh & Patel, 2022). Adopting these systems significantly enhances student information-seeking behavior by offering comprehensive keyword searching and live status updates (Rahman & Lee, 2025). While existing research often focuses on complex AI-assisted cataloging for large-scale universities, a gap remains for smaller institutions. The proposed system addresses this by prioritizing a user friendly interface and real-time availability, avoiding the resource-heavy overhead associated with high-end AI implementations (Chen & Ahmed, 2024).

Automated circulation and transaction management

Automated circulation modules are critical for modernizing library transactions, handling everything from user registration and book loans to overdue notifications in a unified workflow (Gupta, 2020; National Library Network, 2019). Research indicates that these systems significantly improve transaction speed, record accuracy, and user satisfaction compared to manual methods (Iqbal & Mahmood, 2021). While RFID-based systems offer advanced tracking, they often require expensive hardware incompatible with smaller campus budgets (Delos Santos & Cruz, 2022). The system leverages existing camera and mobile technology to achieve these same benefits in speed and efficiency without the prohibitive costs of specialized RFID infrastructure (Lopez & Chandra, 2025).

Inventory management and analytics dashboards

Integrated inventory management systems allow for real-time tracking of collection health and acquisition history, effectively removing the manual labor of stock auditing (Kumar, 2023). When linked with administrative analytics dashboards, these modules enable librarians to monitor usage patterns, such as peak circulation hours and popular resources, transforming library management from reactive to data-driven (Iqbal & Mahmood, 2021; Lopez & Chandra, 2025). The system adopts this approach by providing a dashboard that informs collection development and student recommendations, ensuring library resources are utilized to their full potential based on actual campus demand (Singh & Patel, 2022).

QR-based attendance monitoring

QR code-based attendance systems have emerged as a highly effective, low-cost solution for tracking library occupancy and student flow (Putra et al., 2025). Systematic reviews show these systems reduce administrative time, minimize human error, and provide reliable digital logs for planning and security (Putra et al., 2025; Olanrewaju & Bamidele, 2024). By implementing a QR-based attendance module at STI College Ormoc, the library can provide real-time occupancy data to students while creating an automated history of usage, offering a cost-effective alternative to expensive sensor-based or RFID hardware (Delos Santos & Cruz, 2022; Reyes, 2016).

Online printing service requests

The growth of virtual library services includes the rise of online platforms for supporting student needs, such as off-site printing and copying. Research highlights that moving these requests to a digital platform decentralizes service, clears physical congestion at library counters, and increases operational efficiency (Rahman & Lee, 2025). By embedding an online printing request feature directly into the SmartLib system, the project bridges the gap between digital resource search and physical library services, creating a seamless, self-service experience that accommodates modern student schedules (Olanrewaju & Bamidele, 2024; Rahman & Lee, 2025).

Synthesis

The reviewed literature reveals a significant shift in academic libraries from fragmented, manual record-keeping to integrated, data-driven digital management, demonstrating that automating core operations like cataloging, circulation, and inventory is essential for modernizing library services and enhancing user accessibility. Research consistently shows that integrating emerging technologies, specifically QR-based attendance monitoring and web-based request platforms, provides low-cost, scalable solutions that decentralize services and optimize space, effectively meeting the needs of modern, mobile-dependent students. The proposed SmartLib system aligns with this research trajectory; by unifying these disparate functions into a single platform, it addresses the identified gap in the literature regarding efficient, budget-friendly digital solutions for smaller academic institutions. Ultimately, this project replaces labor-intensive manual processes with a streamlined digital framework, supporting both the administrative objectives of the STI library and the academic requirements of the STI community through evidence-based, smart technology integration.

METHODOLOGY

Technical Background

- **Technologies to be Used**

To develop and implement the SmartLib system, the researchers will utilize a modern tech stack focused on high performance and cross-platform accessibility. The frontend will prioritize user experience through a web application built with HTML, Tailwind CSS, and JavaScript, leveraging React.js to create a highly dynamic and interactive administrative web dashboard. To meet the growing demand for mobile-first solutions, the system will include a dedicated React Native mobile application, providing students and faculty with a seamless, native cross-platform experience for checking in, submitting print requests, and managing reservations directly from their mobile devices. Additionally, an interactive tablet-based local kiosk application built using React.js will be deployed within the library premises for self-service book searches, shelf tracking, and citation generation.

The system's engine and data integrity are powered by a robust backend and secure storage architecture. Node.js with Express.js will handle the core business logic, managing real-time requests for transactions like book borrowing and printing services. Data will be structured and stored using MySQL, optimizing performance by indexing heavily searched attributes such as book titles, authors, and ISBNs to ensure fast querying and support high concurrent requests. To safeguard this data, the researchers will implement a strict Role-Based Access Control (RBAC) hierarchy. This distinguishes three user tiers: Super Admin (Librarian with full configuration access), Library Staff (Student Assistants with restricted operational rights), and users (Students and Faculty). Furthermore, security will be enhanced using dynamic, time-synchronized TOTP (Time-based One-Time Password) QR codes on the student mobile app, which allows secure offline check-in generation while preventing identity fraud through screenshot duplication.

Finally, the system integrates automated technologies, caching, and cloud infrastructure to ensure scalability and operational efficiency. By utilizing barcode scanning for cataloged books and dynamic QR code technologies via tools like html5-qrcode for student identification, the system automates library entry and circulation tracking, significantly reducing manual ledger errors. To handle potential traffic spikes during peak library hours, a local state caching layer will be utilized to minimize redundant database hits. The entire platform will be deployed via cloud-based services like Vercel for the web frontend and Render for the Node.js API, utilizing auto-scaling configurations. To assist in administrative decision-making, the system will feature analytics dashboards built with Chart.js or Recharts, transforming raw data into visual reports on inventory trends, attendance purposes, and printing statistics.

- **Calendar of Activities**

- **Resources**

To successfully develop, test, and implement the SmartLib System, the following hardware and software resources are required:

Hardware Resources

Development Computers: High-performance laptops or desktop computers (minimum Core i5 processor and 8GB RAM) capable of running integrated development environments (IDEs) and local servers.

- Mobile Devices: Smartphones used to test the responsiveness and functionality of the React Native mobile application.
- Barcode and QR Scanners: Dedicated USB barcode scanners or high-resolution webcams used to facilitate and test the automated book-tracking and attendance features.
- Network Equipment: High-speed internet routers and connection devices to support cloud deployment, database synchronization, and real-time notifications.

Software Resources

- Development Environment: Visual Studio Code (VS Code) as the primary code editor, along with the Node.js runtime environment for backend execution.
- Database Management: MySQL for structured data storage and MySQL Workbench for designing and managing database schemas.
- Design and Prototyping Tools: Figma or Canva for creating the User Interface (UI) mockups and mapping the User Experience (UX) flow.
- Version Control: Git and GitHub for source code management, collaborative tracking, and version history.
- Deployment Platforms: Vercel and Render for hosting the web application and backend services in a live environment.
- Documentation Tools: Microsoft Office Suite or Google Workspace for drafting the capstone manuscript, technical manuals, and project reports.
- Web Browsers: Modern browsers like Google Chrome for debugging.

Requirements Analysis

The Requirements Analysis for the SmartLib system identifies the necessary computing solutions to address the specific needs of the institution by evaluating the stakeholders, activities, and existing limitations of the current setup.

The stakeholders involved include the Librarian (acting as Super Admin with full policy control), Library Staff (Student Assistants with restricted operational roles), registered Students and Faculty, and the System Administrator. Regarding the business activity, the system focuses on automating book cataloging with shelf tracking, barcode-based circulation, contactless dynamic QR-based attendance monitoring (incorporating the visitor's purpose of visit), printing request management with format restrictions and live printer status, and a semi-automated GCash receipt validation queue. These functions are designed to operate within the environment of the STI College Ormoc Library premises, while also being accessible via mobile devices, web browsers, and a dedicated tablet-based local kiosk to allow for remote inquiries and self-service.

The timing of these operations primarily occurs during the library's standard hours of 7:00 AM to 5:00 PM, Monday through Saturday, with the system providing real-time updates for due dates and availability based on dynamic operating hours configured by the librarian. Calendar exceptions, including Sundays, official holidays, and school breaks, are dynamically excluded from overdue penalty calculations; if a return date falls on an excluded day, it rolls forward to the next active day. Currently, procedures are performed through labor-intensive manual methods, such as writing in logbooks, searching through card catalogs, and using physical stamps. By replacing these manual workflows with barcode scanning, dynamic QR validation, digital inventory analytics, and automated notification features, the proposed system modernizes operations, ensuring an efficient experience. To manage student accounts over consecutive semesters, a user lifecycle and archiving subsystem is implemented, batch-transitioning graduated or inactive profiles to an 'Archived' status to maintain clean active directories while preserving historic audit logs.

Requirements Documentation

Functional Requirements

The proposed system is divided into two primary interfaces, each equipped with specific features designed to streamline library operations:

User Side Features (Students and Faculty)

- User Authentication: Secure registration and login protocols with account verification to protect user data.

- User Dashboard: A personalized mobile interface displaying currently borrowed books, active reservations, borrowing history, and notifications, alongside shelf location details. The app caches a dynamic, TOTP-encrypted student QR code locally to support time-synchronized offline entry logging.

- Advanced Search and Browse: A multi-tier catalog discovery engine allowing users to query standard literature and academic papers simultaneously by Title, Author, or ISBN. The interface features client-side filtering by Category, Author, Publication Year, and Availability Status. When an item card is selected, the system pulls deep cataloging metadata to display shelf coordinates, real-time availability, and presents an Automated Reference Generator allowing students to view and copy standard APA citations instantly.

- Borrowing and Returning: Digital borrow requests are facilitated by barcode scanning. The system enforces a strict 1-day borrowing period with a due time of 8:59 AM on the next operating day. To allow flexibility, students may request a single-day renewal extension directly from their app before the due time, provided the book has no active reservations and the student has no outstanding fine blocks. The system handles automatic due date monitoring and computes late penalties based on library operating hours.

- Borrowing Constraints and Cart Limits: To ensure equitable resource distribution, the system applies variable cart constraints during the digital reservation and checkout process. Student accounts are subject to a strict borrowing limitation, restricting them to a maximum of two (2) books simultaneously. Faculty and staff accounts are exempted from this restriction and are permitted to add multiple items to their digital cart for academic and instructional use.

- Penalty Computation Rules: The system calculates overdue penalties based on customizable administrative settings (defaulting to ₱2.00/hour and ₱10.00/day starting the following day), up to a configurable maximum penalty cap. Overdue calculation is suspended on library closed days, Sundays, and public holidays, automatically rolling the due date forward to the next operating day.

- Reservation System: The ability to reserve currently unavailable books and join a digital waiting list with automated availability notifications.

- Printing Service Request: A module for uploading files strictly in restricted secure formats (e.g., PDF, DOCX) to prevent malicious scripts, and selecting configurations (copies, color, notes). The system checks printer online/offline status in real time, automatically calculates costs, places the job in the queue, and allows users to track the status. Real-time notifications alert users if a printer is disabled due to stock deficits or maintenance.

- Fine Settlement Options: A module where students can choose to pay their library fines either by scanning the library's GCash QR code (submitting the 13-digit reference number and receipt screenshot for semi-automated verification) or by paying in cash directly at the library front desk.

- QR Attendance Monitoring and School ID Integration: A contactless entry and transactional verification system supporting dual authentication: the dynamic, mobile app QR code and the static QR code on student ID cards. This integration automates attendance logging and verifies student identities for library desk transactions (such as borrowing and returning books) as well as self-service functions.

- Tablet Kiosk Interface: A dedicated local tablet web interface allowing students to perform rapid advanced catalog searches, view book information, identify shelf locations, and copy APA citations instantly within the library.

- Notification System: Automated alerts for due dates, overdue penalties, reservation arrivals, and printing updates.

- Digital Clearance Status: A dedicated module where users can view their real-time library clearance standing (e.g., "Cleared" or "Not Cleared"). If a student is blocked, the system explicitly displays the specific unreturned books or outstanding penalty fines causing the hold.

Librarian and Admin Side Features

- Dashboard and Analytics: A centralized administrative dashboard tracking Key Performance Indicators (KPIs) including Total Books, Active Borrowed Books, Most Borrowed Books, Active Users, Daily Attendance (segmented by purpose of visit), Available Books, Overdue Books, Active Reservations, Total Penalties, Recent Activities, Book Availability Stats, and Returned Books.

- Book and Research Management: Full CRUD (Create, Read, Update, Delete) capabilities for books and thesis materials, including barcode assignment and category management.

- Category Management Sub-Table: The category interface must feature an explicit data grid displaying Category ID, Category Name, Description, Number of Books currently tied to that category, and Date Added, alongside direct triggers for Edit Category and Delete Category actions.

- Research and Thesis Inventory Analytics: The research repository module must feature a dedicated data-visualization component capable of rendering a Department Analysis (aggregating and grouping submitted manuscripts by their respective academic programs) and Year Trends (tracking and graphing the volume of institutional research submissions across consecutive graduation years)

- Transaction Monitoring: A centralized hub to track all borrowing cycles, approve requests, monitor overdue items, and export detailed transaction reports.

- Fines Management Hub: A dedicated administrative workspace providing segmented visibility into penalties. This module displays paid/unpaid fines, maps book and user data, provides custom actions to issue non-circulation infraction fines (e.g., noise, eating), and handles payment processing. For online transactions, it features a GCash Verification Queue to inspect uploaded receipts and reference numbers. For manual transactions, it provides an instantaneous 'Cash Payment' trigger, allowing librarians or staff to settle fines and clear student records directly at the front desk, exporting reports for accounting audits.

- Inventory Management: Automated inventory audits using barcode scanning, tracking of missing or damaged resources, and comprehensive inventory reporting.

- Printing Service Management: Tools to manage print queues, toggle printer online/offline states, log current ink/paper stock levels, update job statuses, and track revenue and usage statistics.

- Printing Supplies Asset Management: A dedicated inventory and analytics layer tracks physical library printing consumables in real time. The ink repository submenu tracks unique Ink IDs, ink cartridge types, color variations, remaining fluid levels, and specific printer assignments, throwing low-ink triggers when thresholds are breached. The system auto-triggers printer offline status warnings when consumables reach zero. The bond paper stock matrix monitors paper size dimensions (e.g., Short, A4, Long), remaining reams, replenishment histories, and expense trends to generate data-driven supply consumption reports for administrative budgeting.

- Attendance and User Management: Real-time monitoring of QR logs, generation of attendance reports, and the ability to manage user roles or account statuses.

- User State Lifecycle Actions: The administration table provides triggers to

Activate, Deactivate, or Archive user profiles. The archiving action transitions graduated or inactive accounts to an archived state, disabling system login while keeping their transaction records intact for historical reporting.

- Role-Based Access Control (RBAC): Enforces permissions separating the Librarian (Super Admin with full settings, fine modifications, and archiving control) from Student Assistants (restricted staff access for borrowing scans, print queues, and GCash approvals, with no deletion or policy settings rights).
- Bulk Data Import: An administrative tool allowing the librarian to upload files to bulk-register students or import large inventories of books into the database.
- Automated Clearance Management: A master dashboard that automatically flags students as "Cleared" or "Pending" based on their transaction history. It allows librarians to search the student master list, view individual clearance blocks, manually override statuses for special cases, and export bulk clearance reports for campus submission.

Design of Software, System, Product, and/or Processes

To clearly map the operational transition from legacy procedures to the proposed digital ecosystem, this section presents three architectural workflows. Figure 1 (Manual Attendance Process) establishes the 11-step paper logbook path currently required for facility entry. Figure 3 (Manual Process of Printing) chronicles the physical medium vulnerabilities inherent to legacy file handling and cash recording. Figure 3 (Design Process) charts the unified software system architecture, outlining student access privileges, search indexing logic, dynamic automated queues, and immediate notifications.

MANUAL PROCESSES

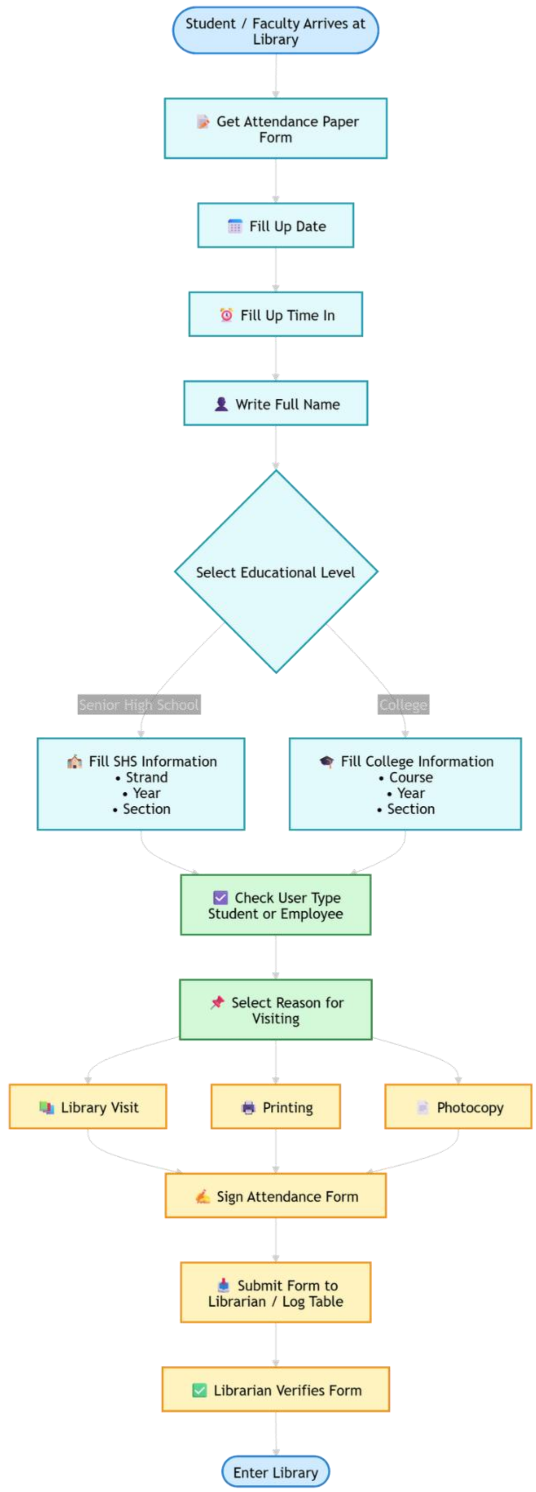

Figure 1. Manual Process Attendance

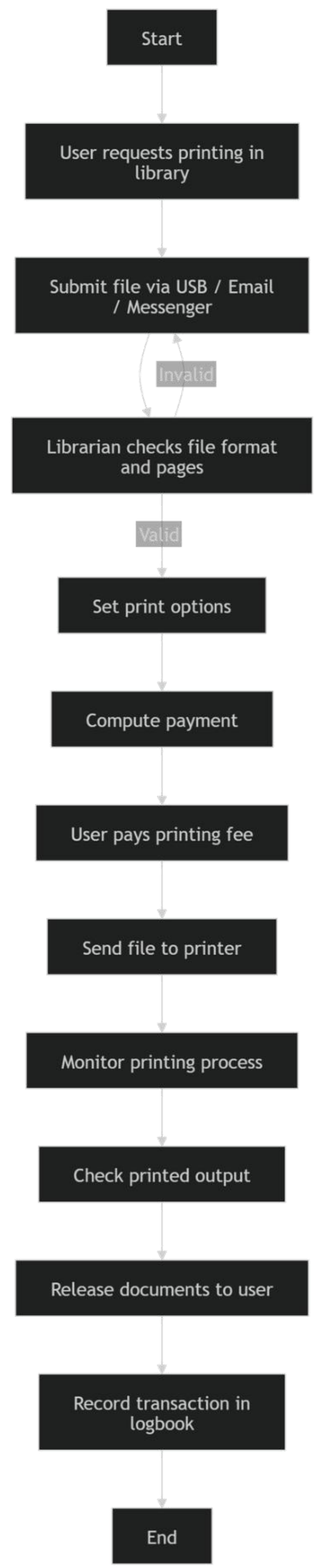

Figure 2. Manual Printing Process

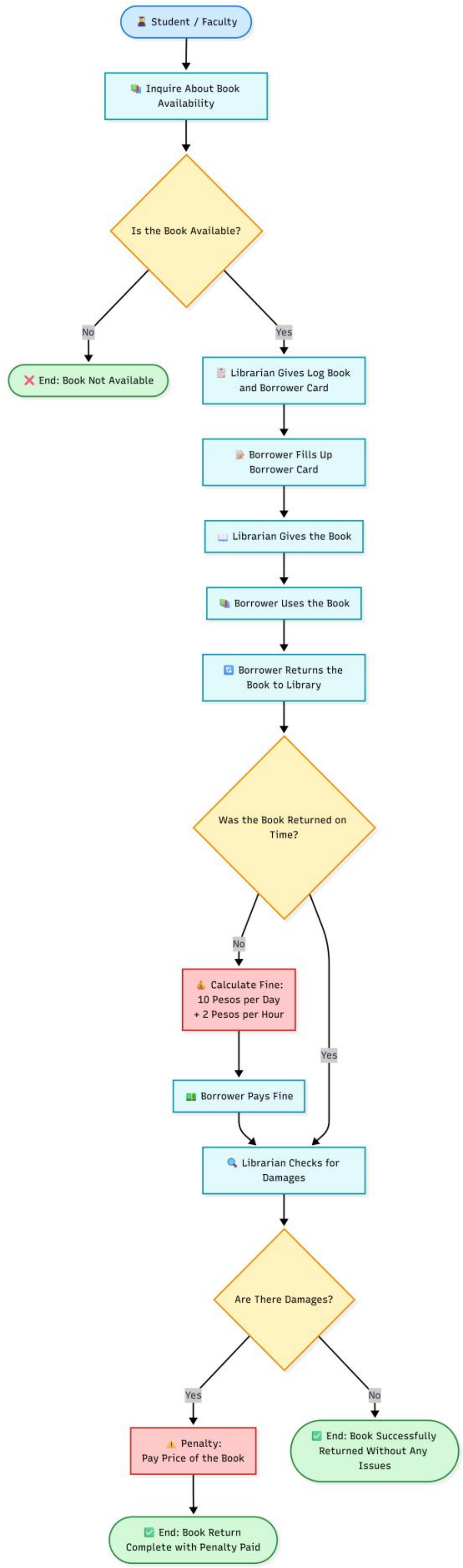

Figure 3. Manual Borrowing Process

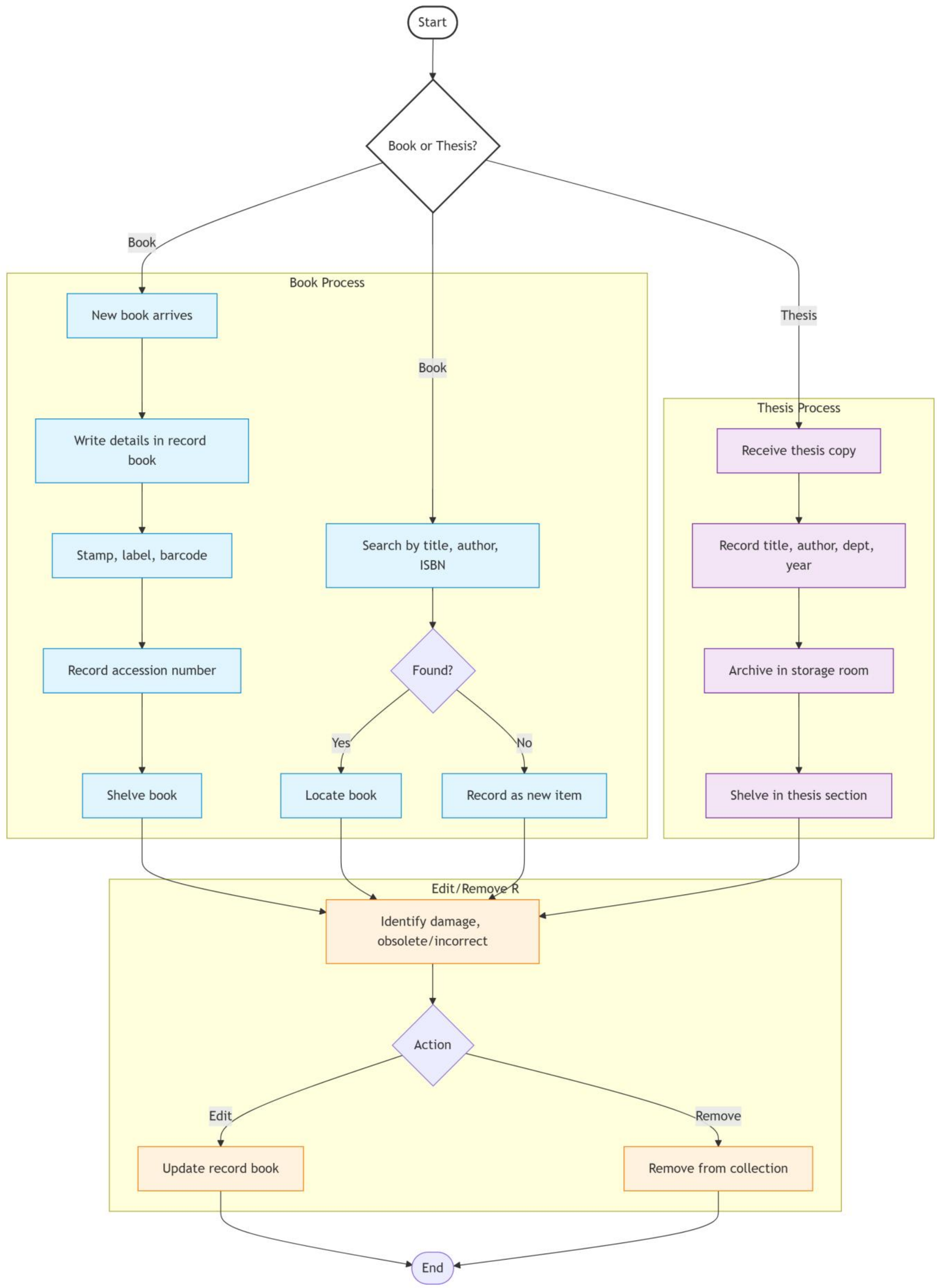

Figure 4. Manual Books Management Process

ADMIN SIDE

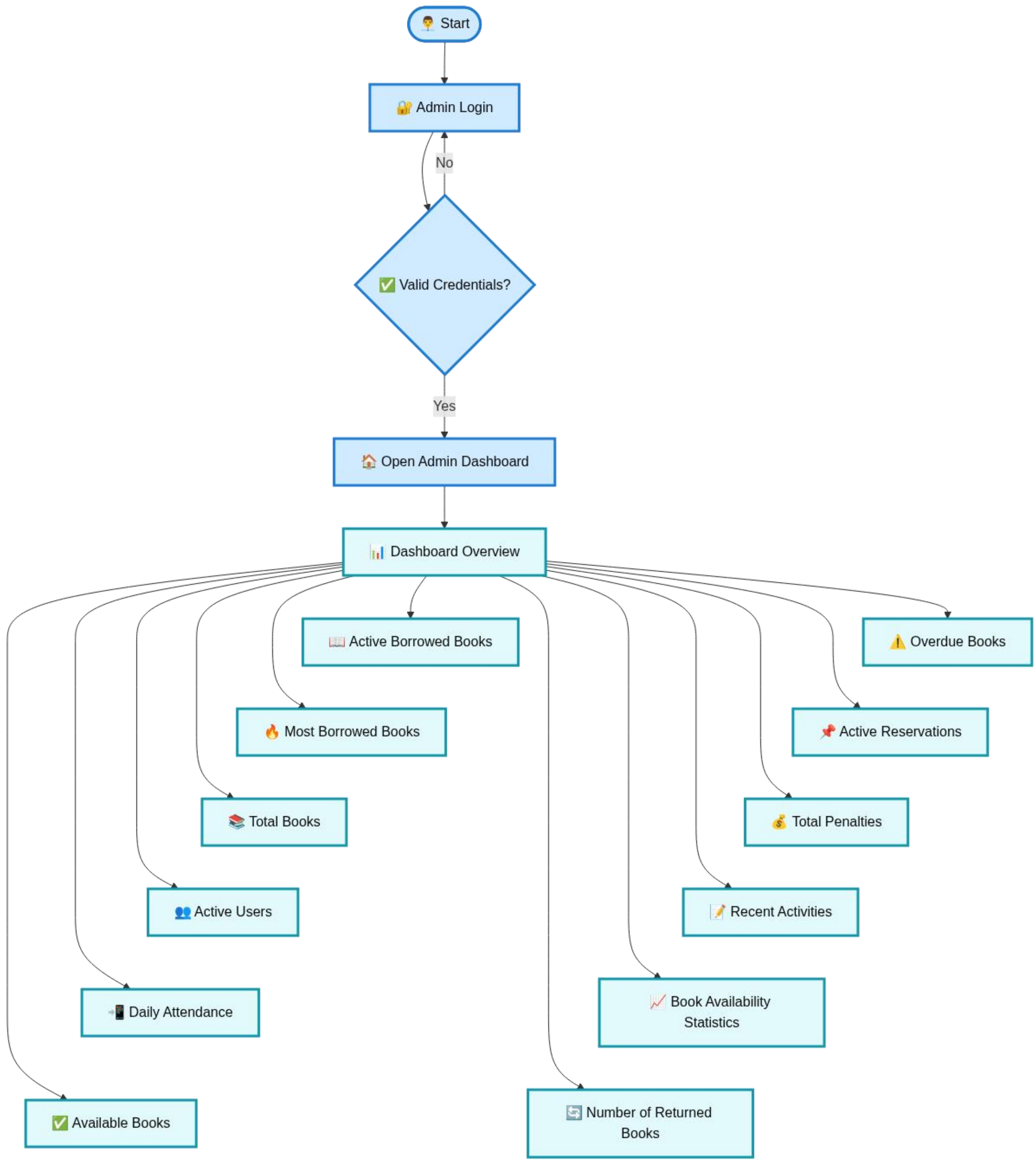

Figure 5. Proposed Process Dashboard (Admin Side)

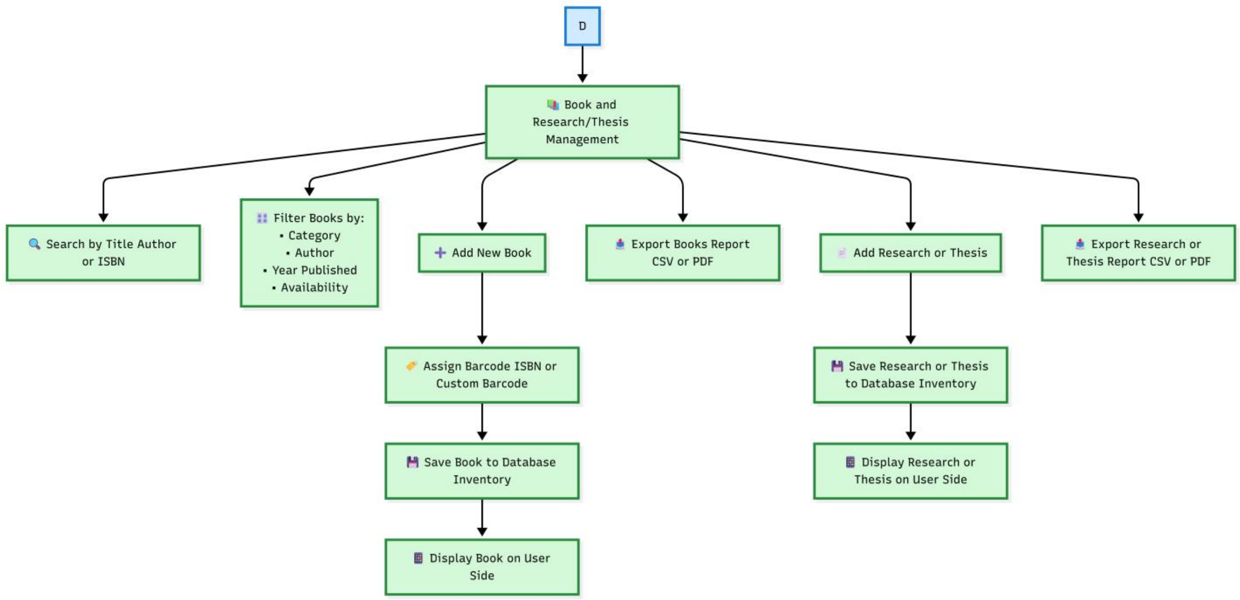

Figure 6. Proposed Book Management Process (Admin Side)

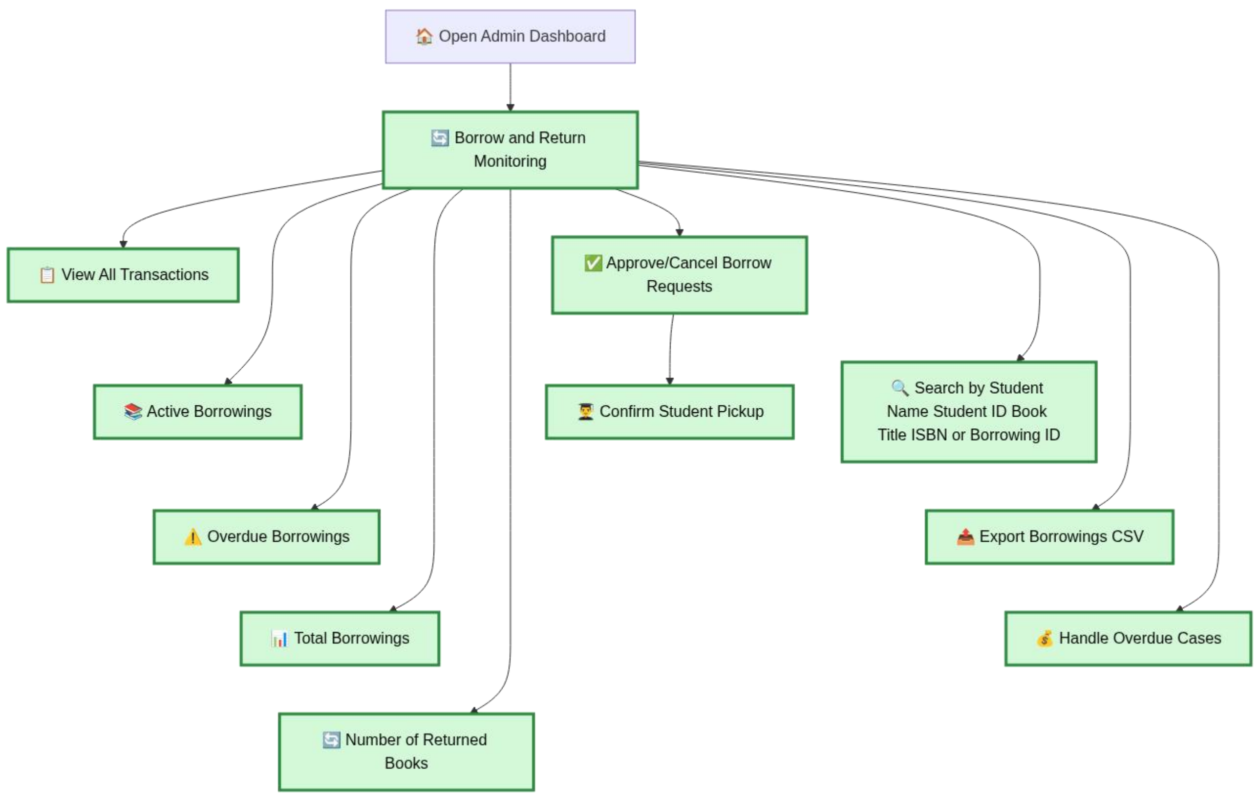

Figure 7. Proposed Borrow and Return Monitoring Process (Admin Side)

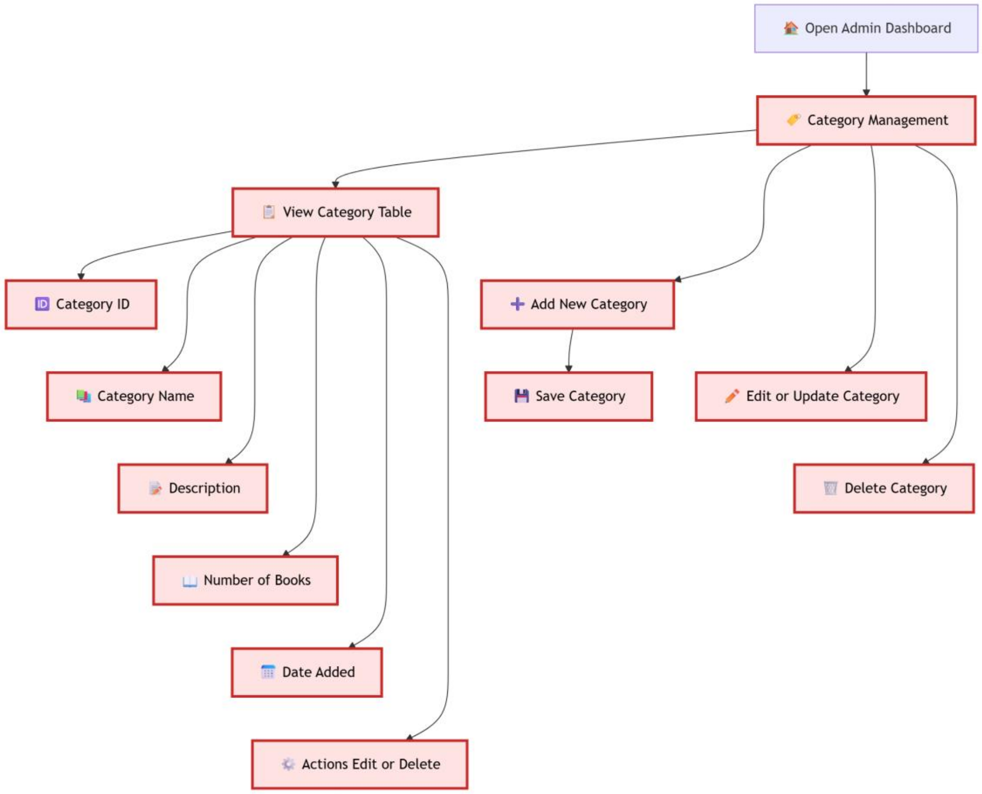

Figure 8. Proposed Category Management Process (Admin Side)

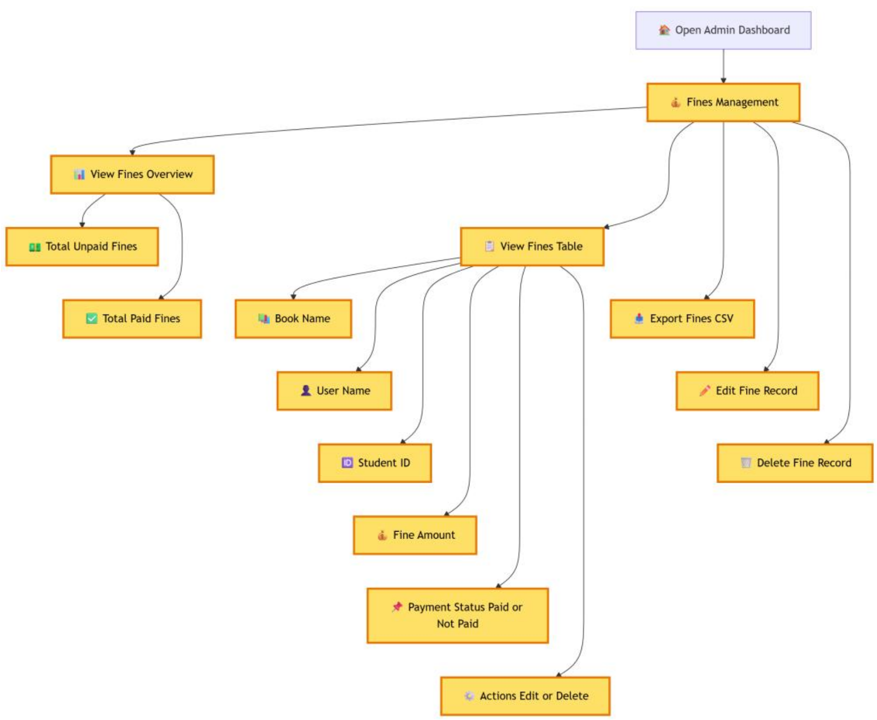

Figure 9. Proposed Fines Management Process (Admin Side)

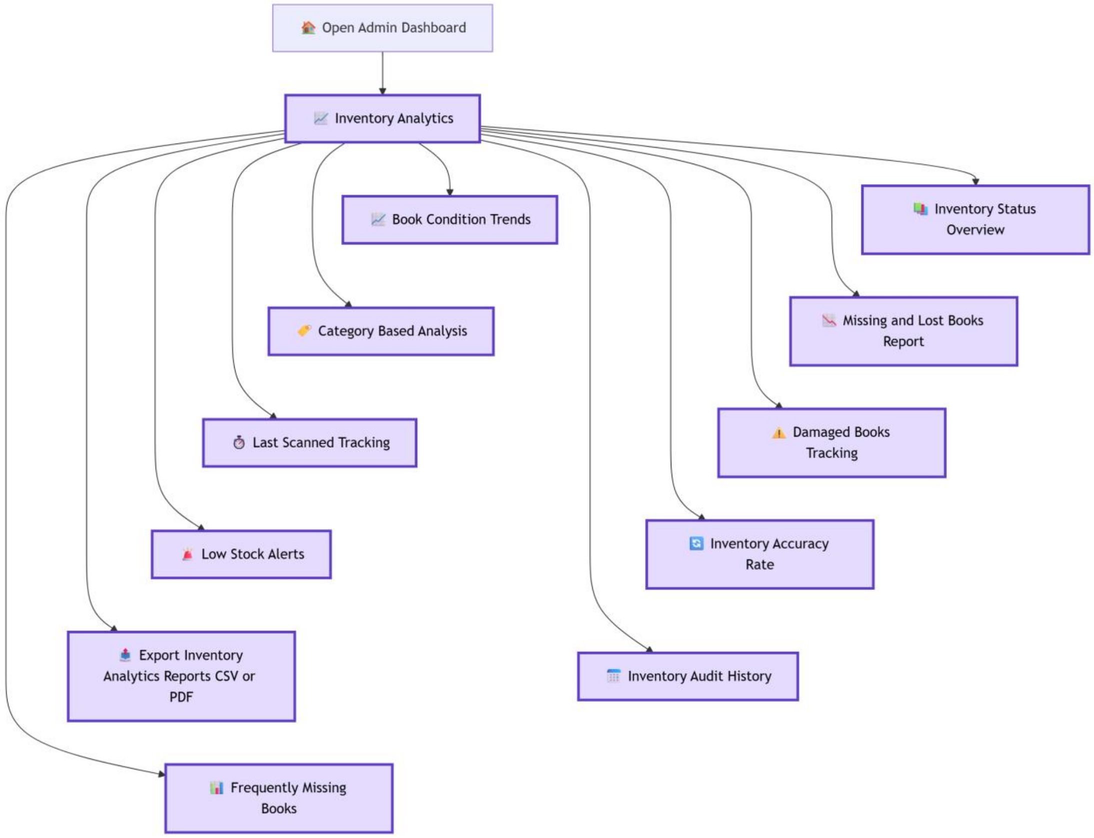

Figure 10. Proposed Inventory Analytics Process (Admin Side)

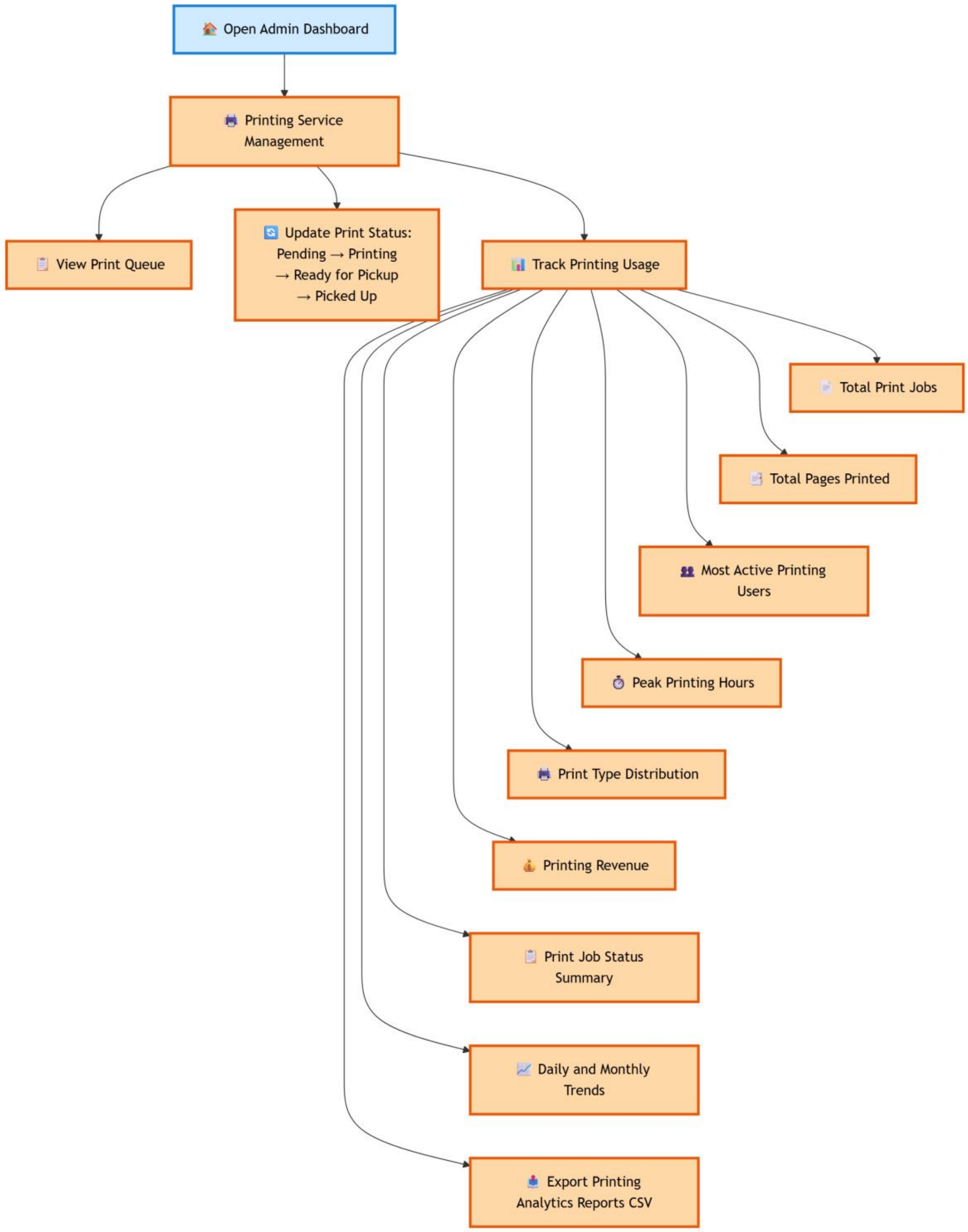

Figure 11. Proposed Printing Service Management Process (Admin Side)

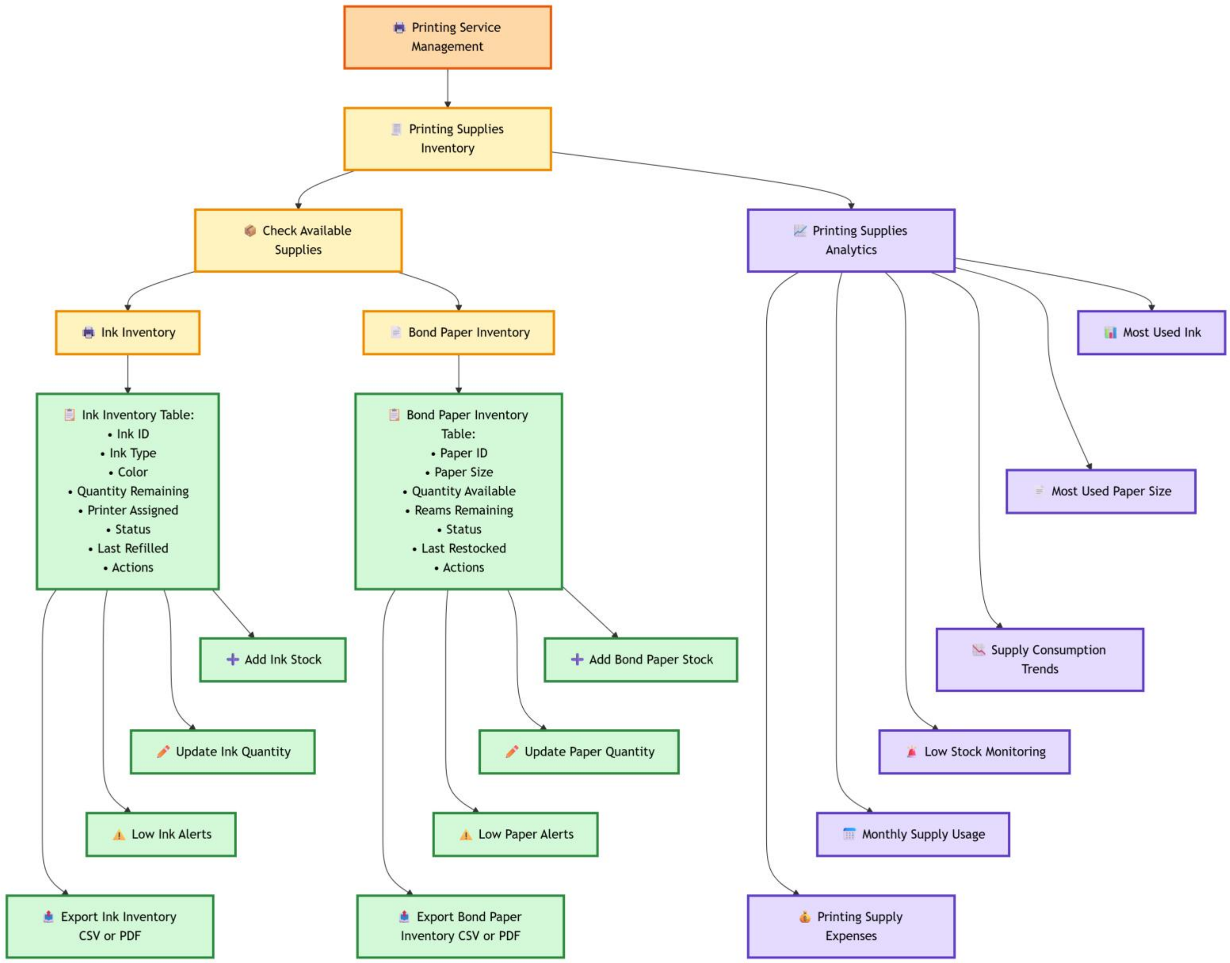

Figure 12. Proposed Printing Supplies Inventory/Analytics Process (Admin Side)

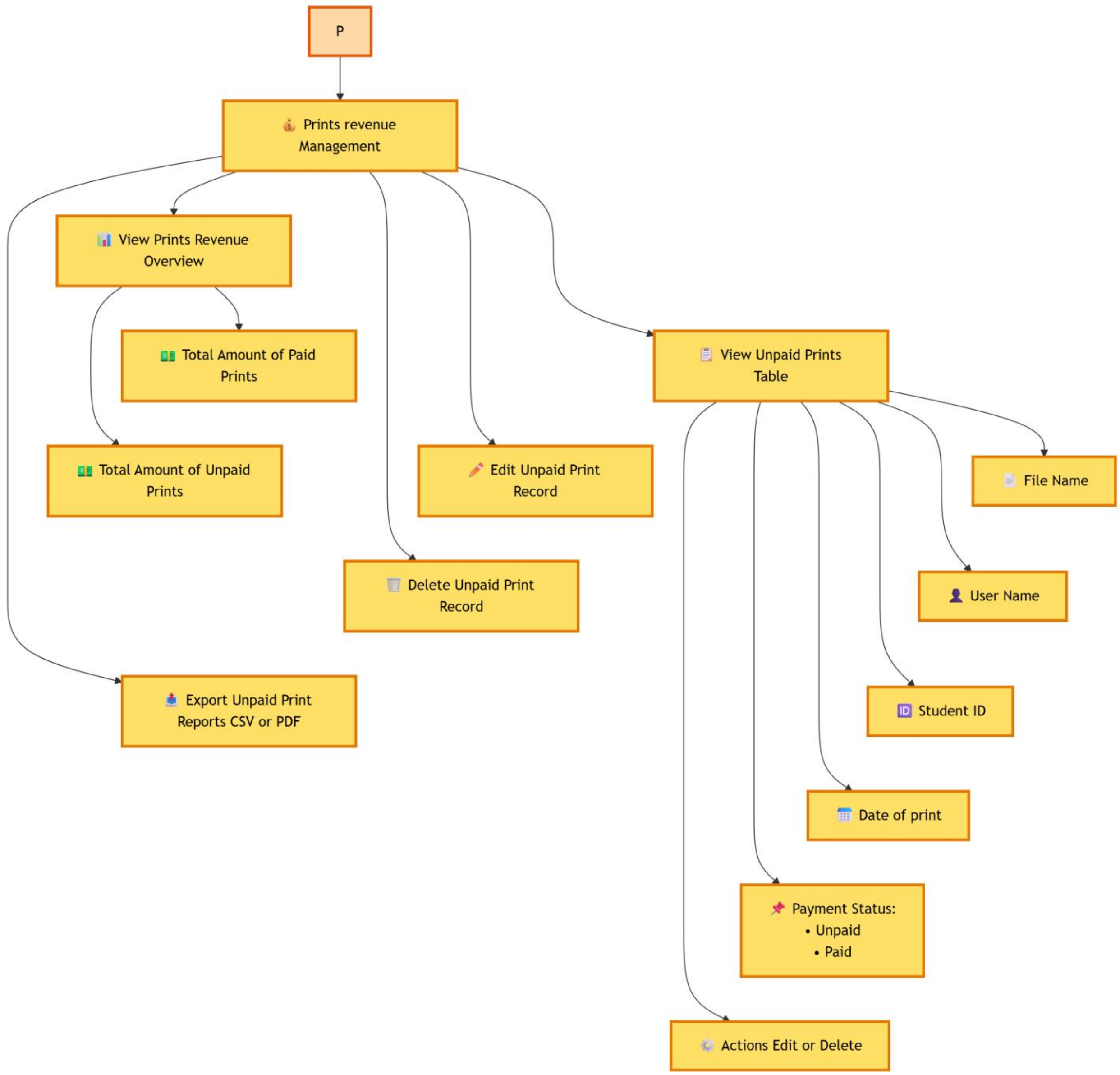

Figure 13. Proposed Prints Revenue Management Process (Admin Side)

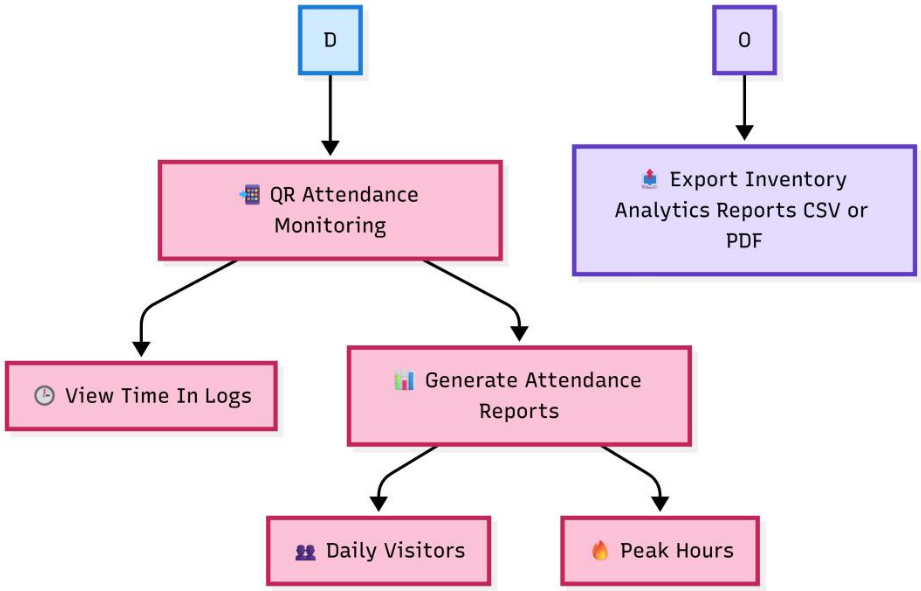

Figure 14. Proposed Attendance Monitoring Process (Admin Side)

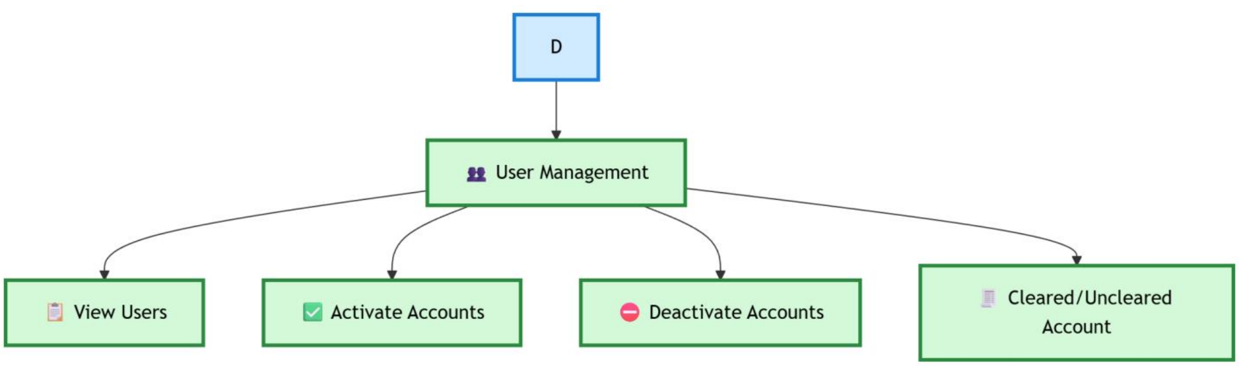

Figure 15. Proposed User Management Process (Admin Side)

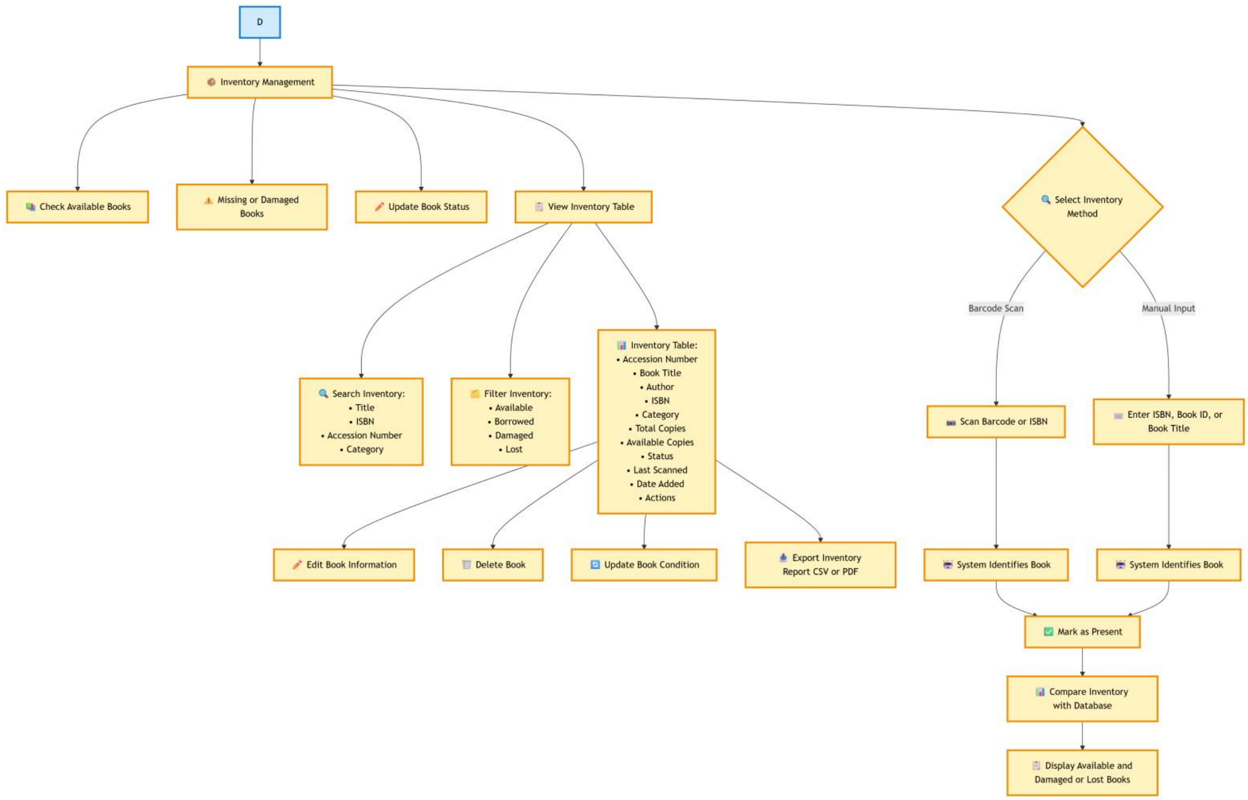

Figure 16. Proposed Inventory Management Process (Admin Side)

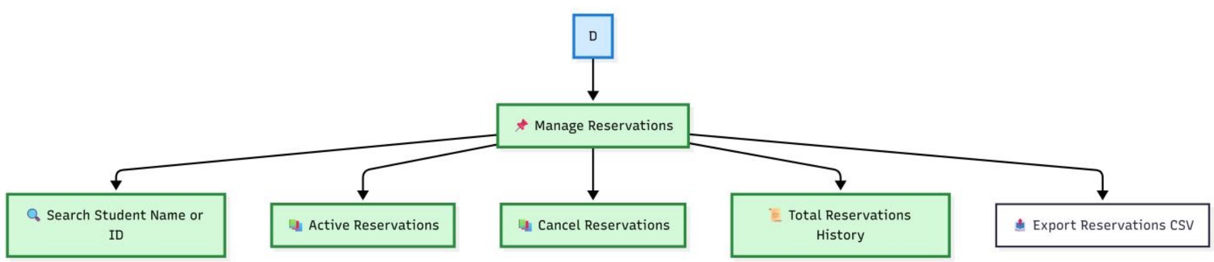

Figure 17. Proposed Reservation Management Process (Admin Side)

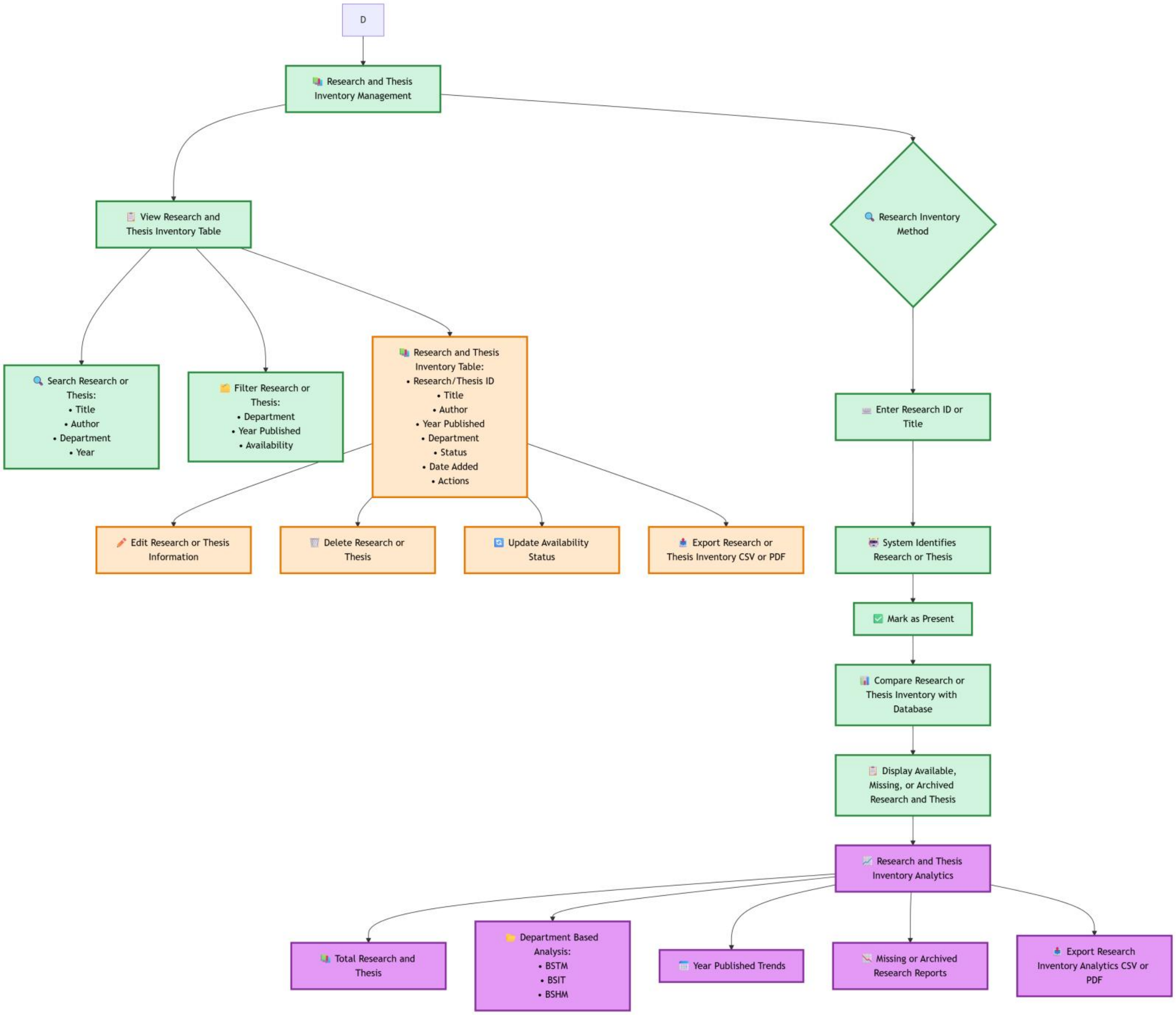

Figure 18. Proposed Research and Thesis Inventory Management Process (Admin Side)

USER SIDE

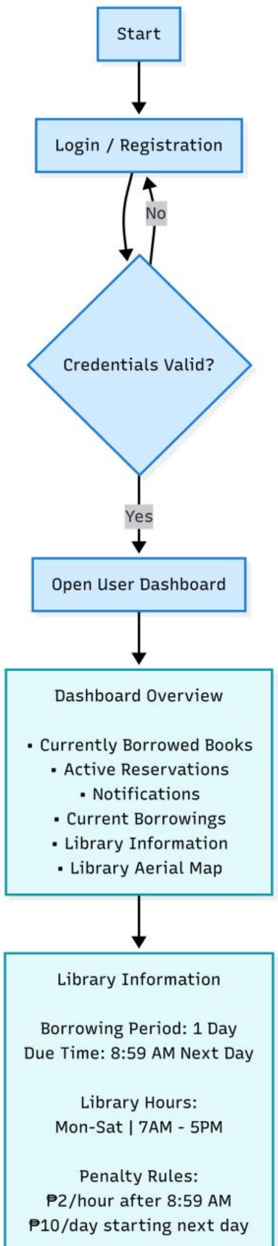

Figure 19. Proposed User Dashboard Process (User Side)

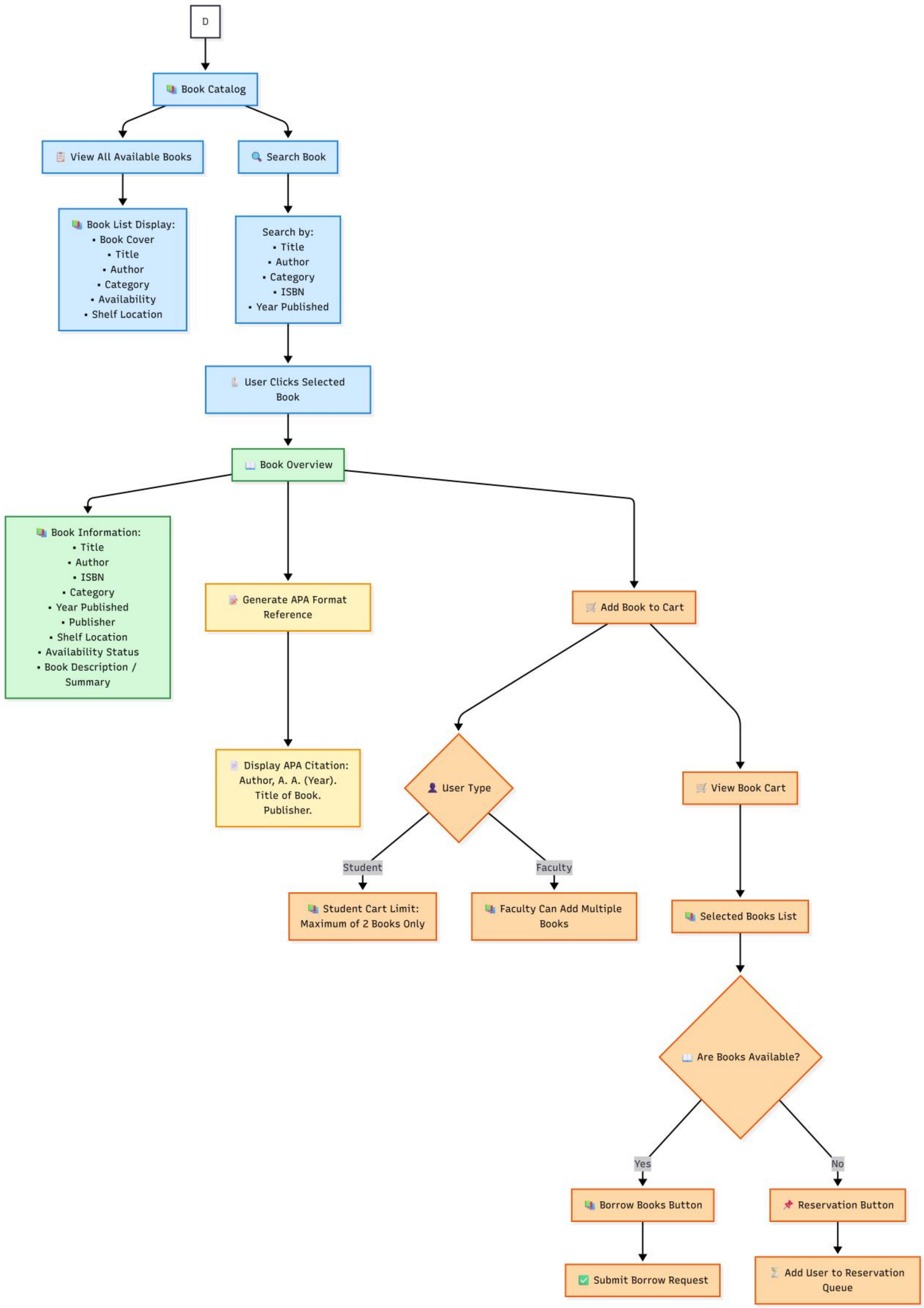

Figure 20. Proposed Book Catalog Process (User Side)

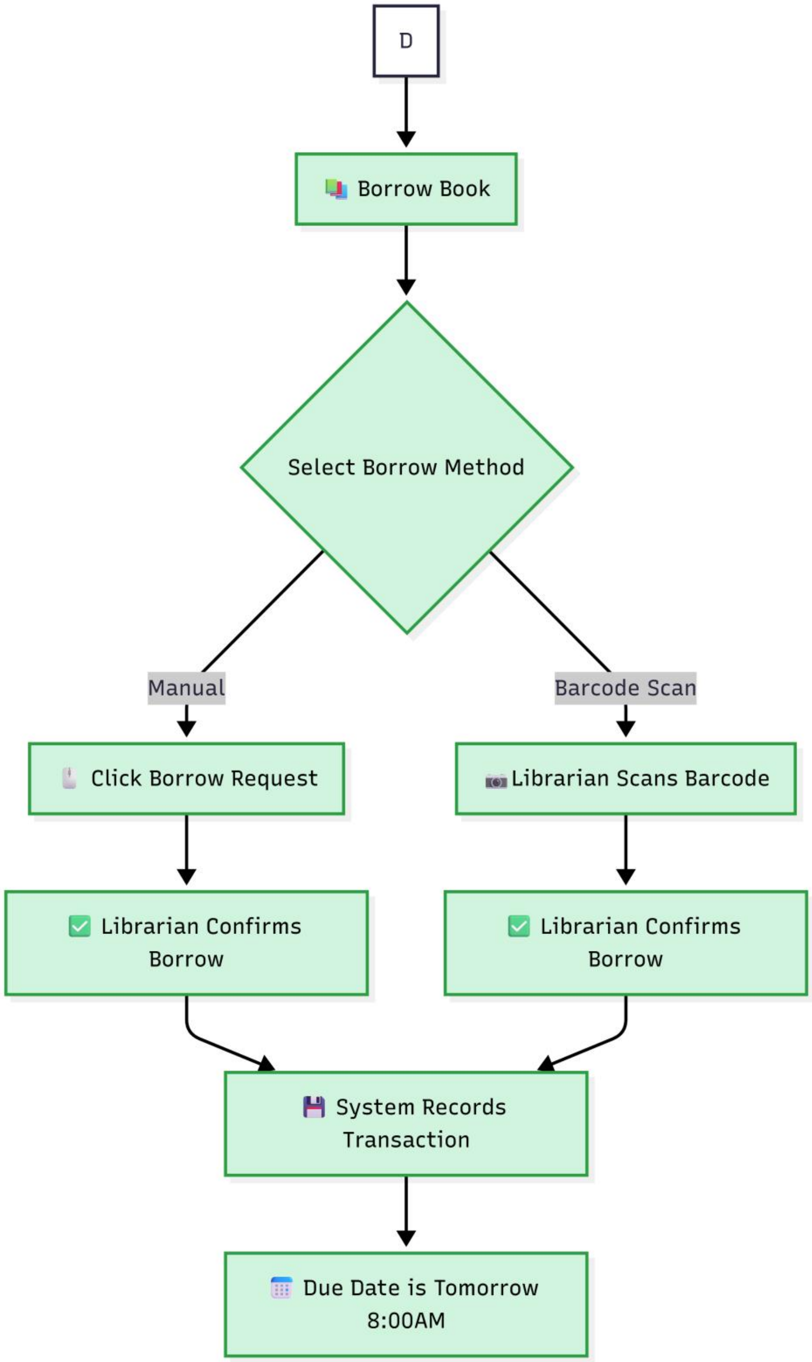

Figure 21. Proposed Borrowing Process (User Side)

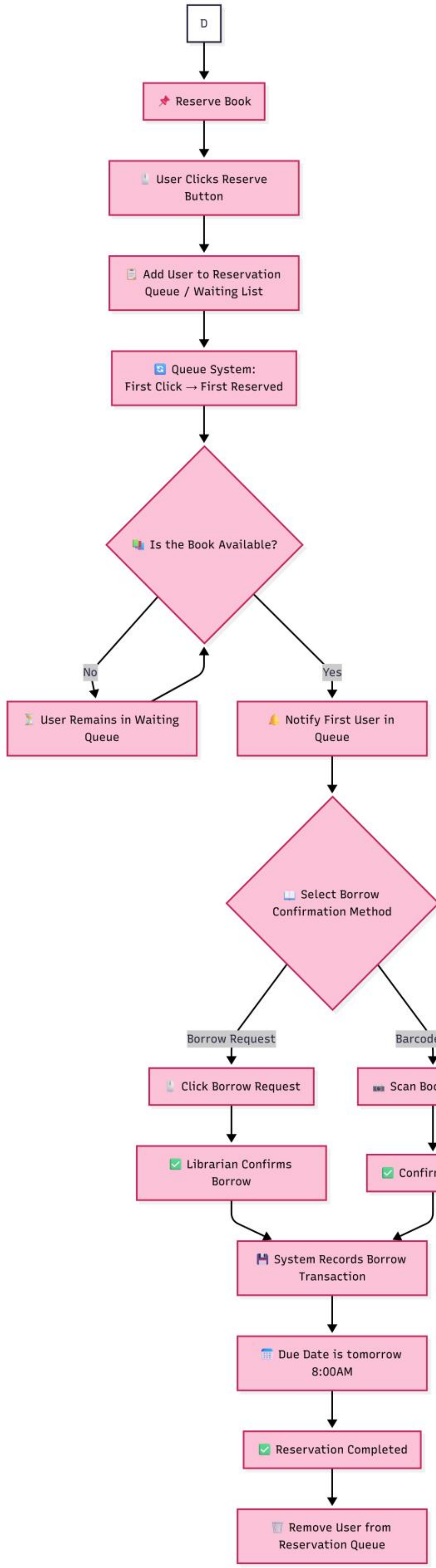

Figure 22. Proposed Reservation Process (User Side)

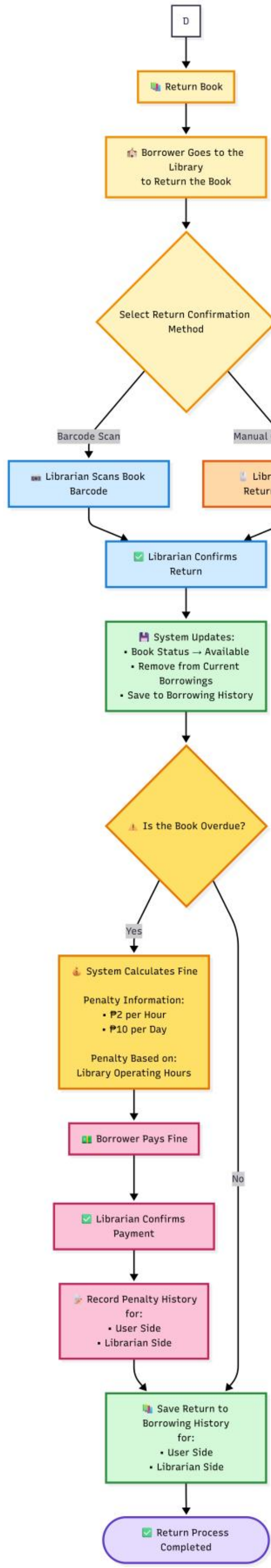

Figure 23. Proposed Returning Process (User Side)

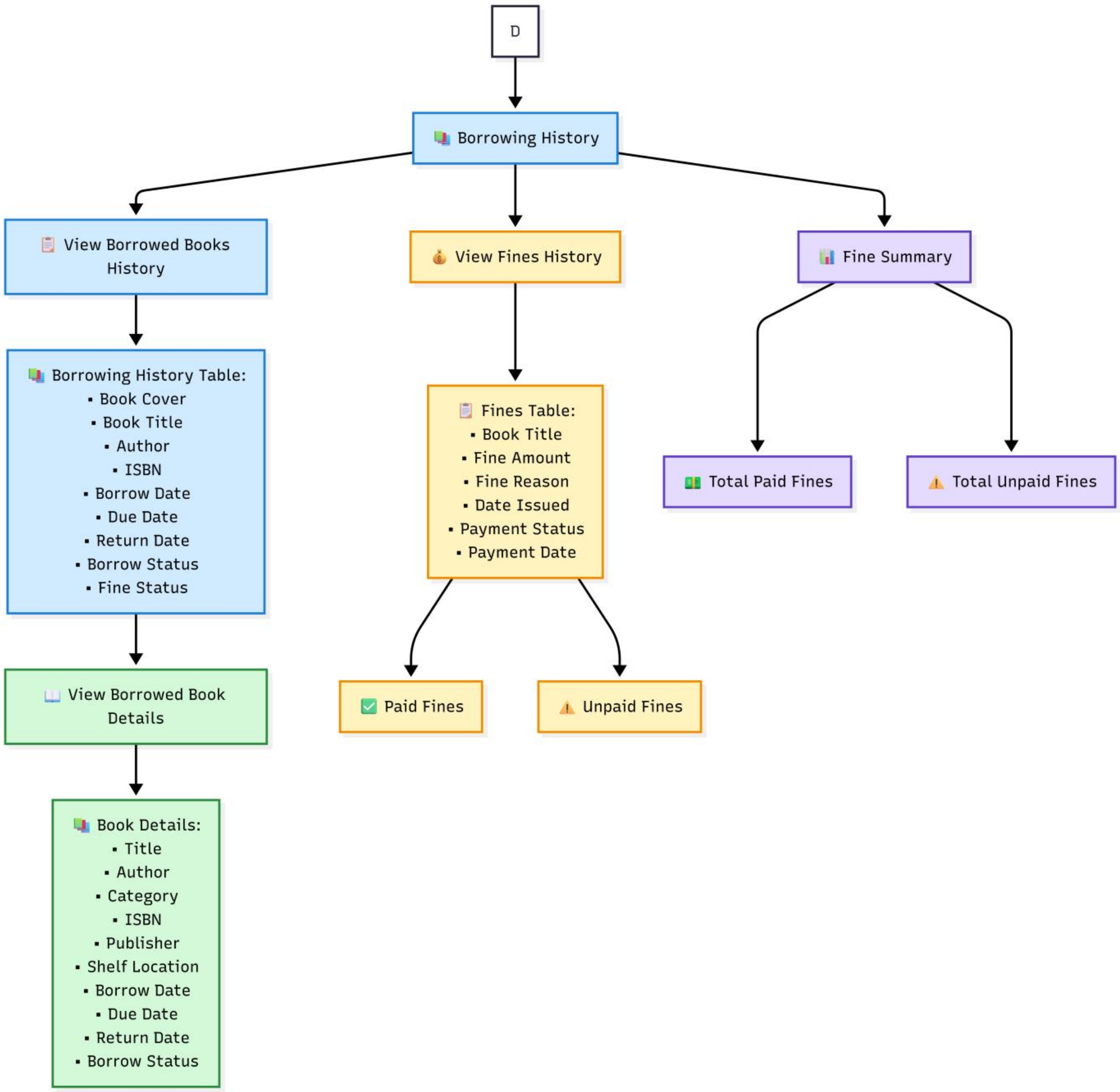

Figure 24. Proposed Borrowing History Process (User Side)

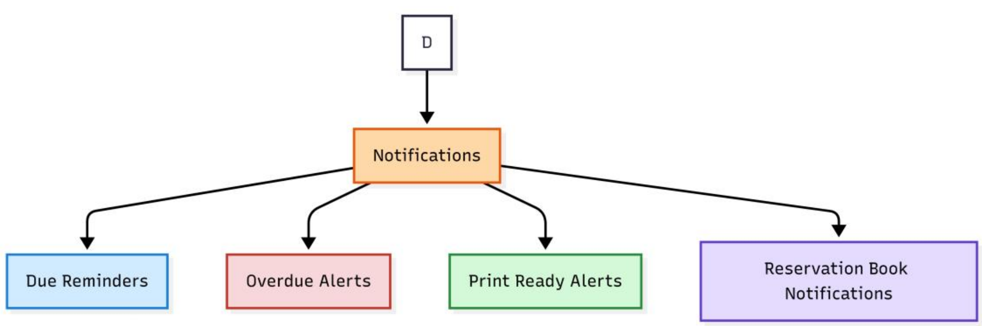

Figure 25. Proposed Notifications Process (User Side)

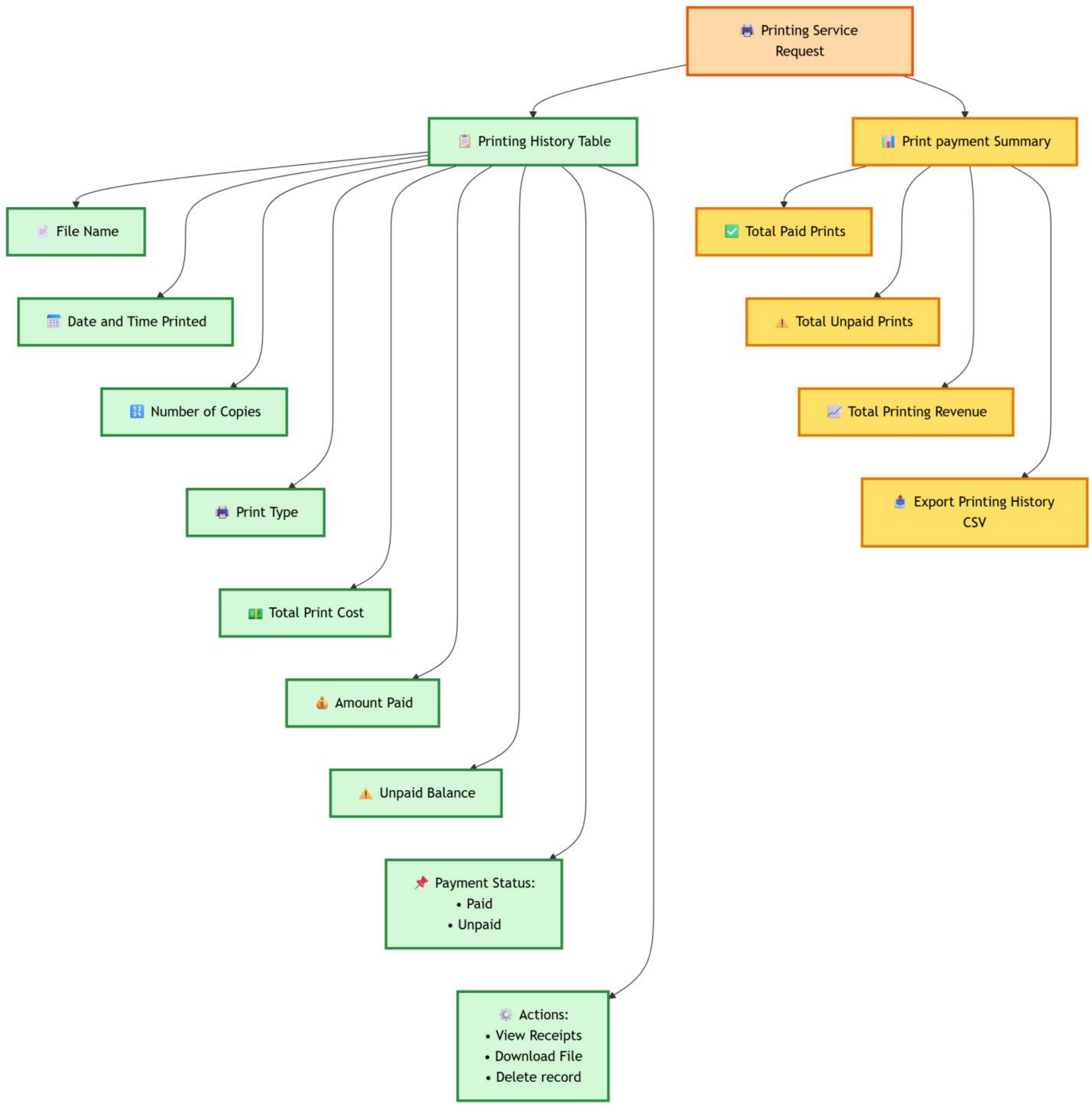

Figure 26. Proposed Print History and Print Payment Summary Process (User Side)

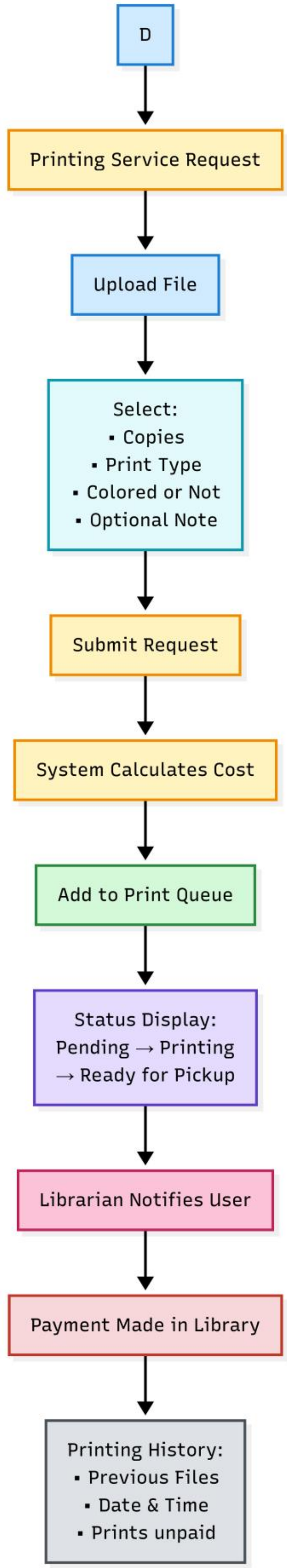

Figure 27. Proposed Printing Service Request Process (User Side)

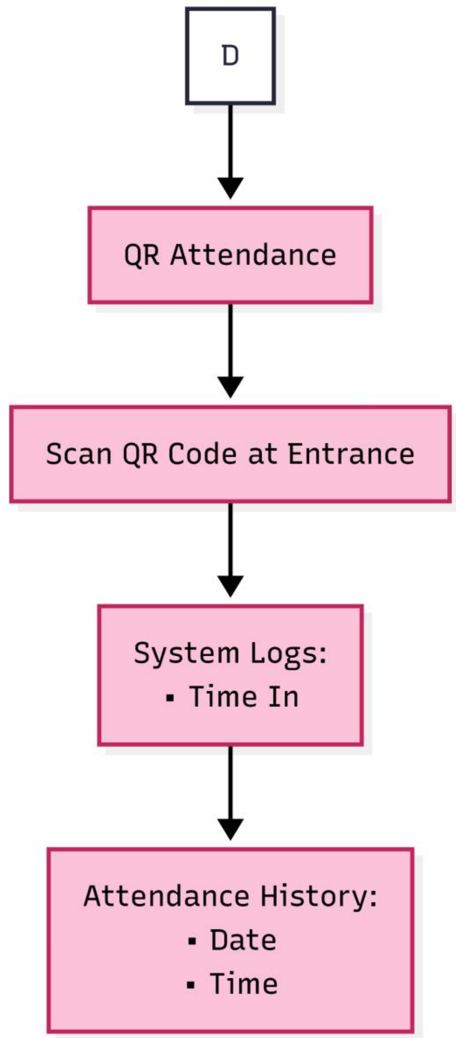

Figure 28. Proposed QR Attendance Process (User Side)

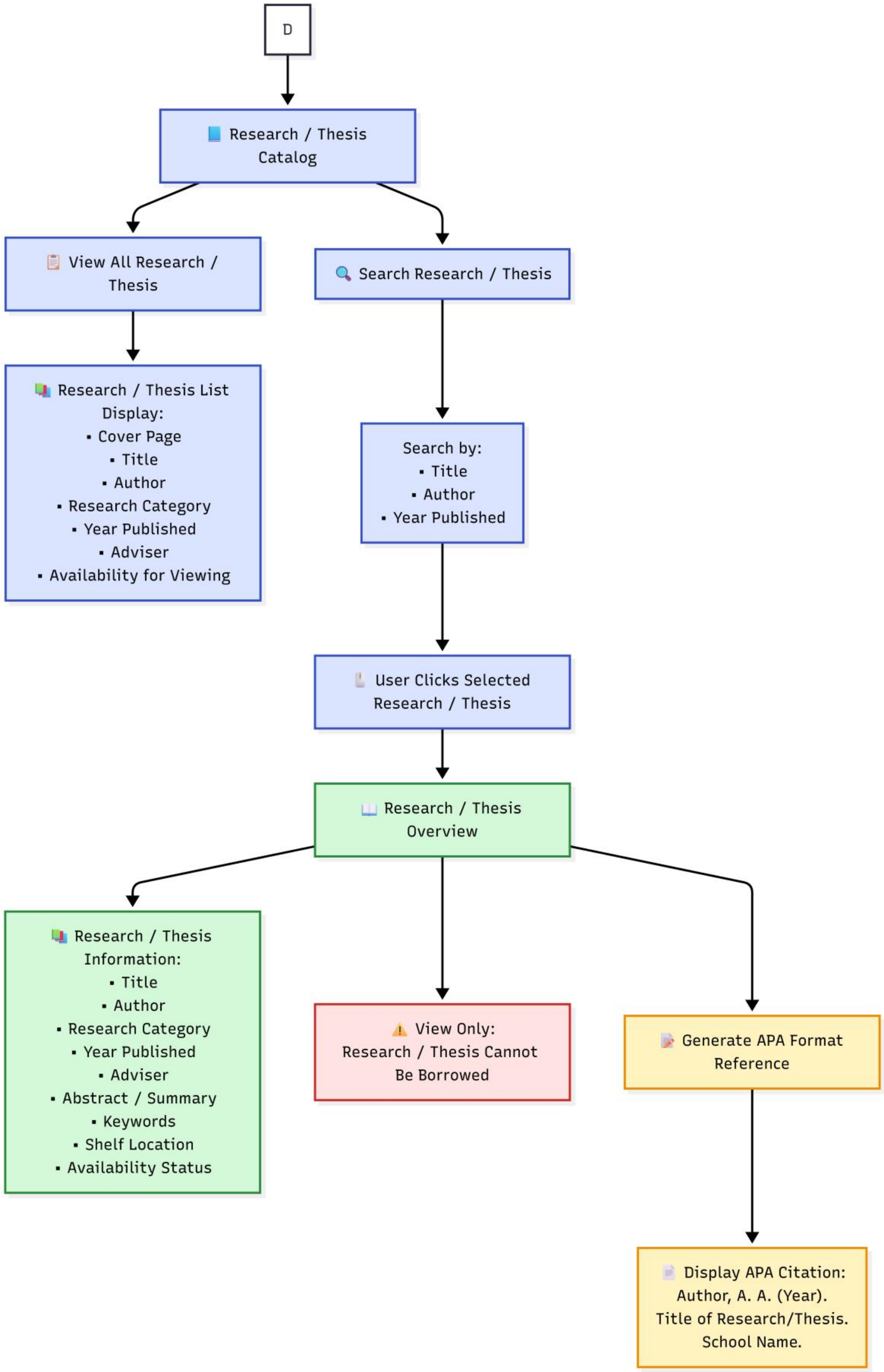

Figure 29. Proposed Research/Thesis Catalog Process (User Side)

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APPENDIX A. RESOURCE PERSONS

Engr. Sheena Joy S. Muyuela
Profession: Academic Head
Name of Institution: STI College Ormoc

Rona Mira B. Lucañas
Profession: IT Program Head / Capstone Project Coordinator
Name of Institution: STI College Ormoc

Elizabeth L. Dumaran
Profession: IT Instructor / Capstone Project Adviser
Name of Institution: STI College Ormoc

Judelyn Martinez
Profession: Librarian
Name of Institution: STI College Ormoc

APPENDIX B. PERSONAL TECHNICAL VITAE

Curriculum Vitae of

JOHN BAZTY C. CANTAY

BARANGAY TUGBONG, SITIO PUNAY KANANGA LEYTE

cantay242@gmail.com

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-2027	STI College Ormoc
Vocational/Technical	2020-2023	KNHS Senior Highschool
High School	2016-2020	Kananga National Highschool
Elementary	2011-2016	Cacao Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
	N/A	N/A

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	IT Club	Member
2025	IT Club	Member
2024	IT Club	Member
2023	IT Club	Member

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
HTML	Entry	September 2025
CSS	Entry	September 2025
UI/UX Design	Intermediate	November 2025
Node.js	Entry	March 2026

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
September 2025	PayTaca Workshop
December 2024	National Service Training Program

Listed in reverse chronological order (most recent first).

Curriculum Vitae of

KEITH GAUDI GERARD P. JUNTILLA

BRGY. LIBERTAD, ORMOC CITY, LEYTE

juntilla.keith@gmail.com

09602700364

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2025 - Current	STI College Ormoc
Vocational/Technical	2018 - 2021	Saint Peter's College of Ormoc
High School	2014 - 2018	Saint Peter's College of Ormoc
Elementary	2008 - 2014	Saint Peter's College of Ormoc

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	IT Club	Member
2025	IT Club	Member

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
Web Development (HTML & CSS)	Basic	January 2026
Android Studio (Mobile Dev)	Basic	January 2026
C, C#, Java Programming	Basic	August 2025

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
April 2026	Research Seminar: Bridging Academe & Industry through Research
November 2025	Cybersecurity: Digital First Responders - From Evidence to Ethics Seminar

Listed in reverse chronological order (most recent first).

Curriculum Vitae of

BUENTHEO NATHANIEL A. NOVAL

BARANGAY PUNTA, ORMOC CITY LEYTE
ethannoval99@gmail.com

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-2027	STI College Ormoc
Vocational/Technical	2020-2023	STI College Ormoc
High School	2016-2020	NOCNHS
Elementary	2011-2016	Ormoc City Central School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2022-2023	Graphics Designer	WELLIFE

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	IT Club	Member
2025	IT Club	Member
2024	IT Club	Member
2023	IT Club	Member

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
HTML	Intermediate	January 2021
CSS / Tailwind CSS	Intermediate	January 2021
Javascript	Intermediate	January 2023
Frontend debugging	Intermediate	March 2024

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
September 2025	PayTaca Workshop
December 2024	National Service Training Program

Listed in reverse chronological order (most recent first).

APPENDIX C. TRANSMITTAL LETTER

May 4, 2026

Ms. Judelyn Martinez
Librarian
STI College – Ormoc

Subject: Request for Requirements Gathering and Workflow Consultation

Dear Ma'am,

Good day.

Following our initial proposal for the Automated Library Management System (LMS), we, the researchers, are now entering the critical phase of requirements gathering. To ensure that the final system effectively addresses the specific needs of our campus, we respectfully request a short window of your time today, May 4, 2026, to conduct a brief interview.

Our primary objectives for this visit are:

- To gain a deeper understanding of the library's day-to-day operations and current administrative workflows.
- To further discuss and understand the desired functionalities of the system's features to ensure they provide maximum utility to both staff and students.
- To identify opportunities where digital automation can best support the library's staff and improve service delivery to students.
- To gather feedback on proposed system functionalities and ensure they integrate seamlessly with the library's long-term goals.
- To discuss the specific data management and reporting requirements essential for efficient library oversight.

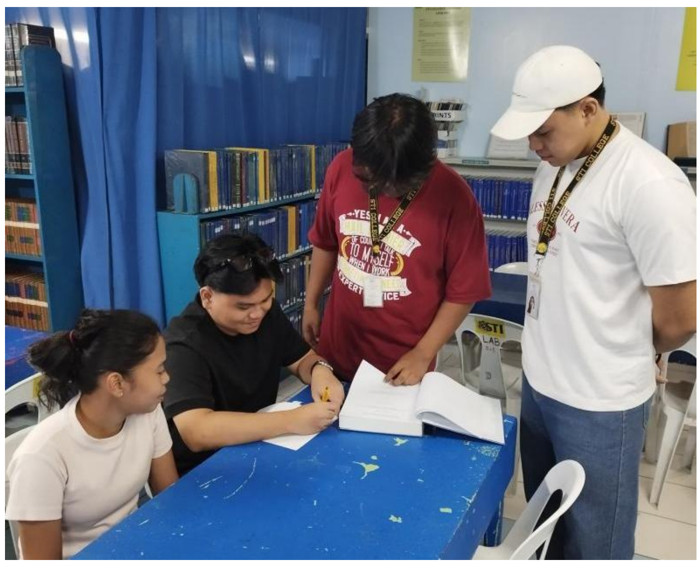
We understand that your schedule is busy and will ensure our consultation is concise and focused. Our goal is to collaborate closely with you to build a system that minimizes manual workload and enhances the library experience for all STIers.

Thank you for your continued support and guidance in this project. We look forward to your favorable response.

Sincerely yours,

John Bazty C. Cantay
Keith Gaudi Gerard Juntilla
Buentheo Nathaniel A. Noval

APPENDIX D. PROJECT DOCUMENTATION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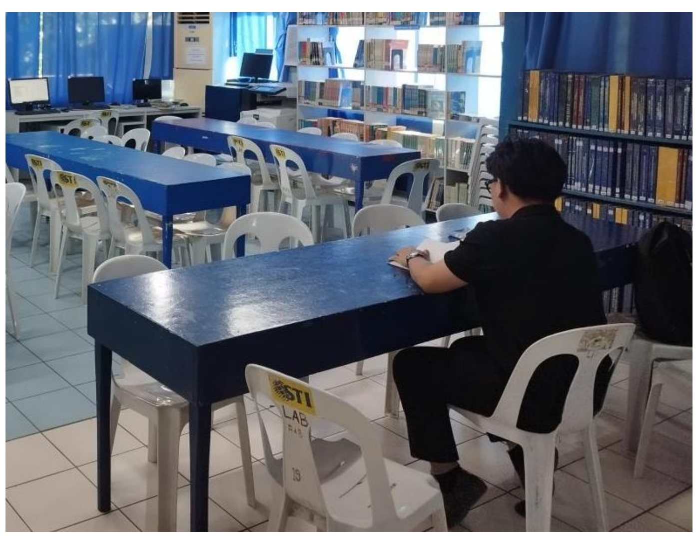
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